A Local Theory of Gradient-Based Rank-One Recovery in Nonlinear Low-Rank Models

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Abstract

In linear low-rank recovery, the next rank-one component is read off from the current residual matrix; with a nonlinearity in place, the residual is hidden and only a gradient is observed. Motivated by sequential rank-one LoRA, we study a direction-locked sequential rank-one recovery procedure in a single-layer nonlinear teacher–student model. At each stage k, the procedure picks a rank-one direction $\hat{u}_k\hat{v}_k^\top$ from the gradient of the squared-error adaptation loss $\mathcal{L}_N$ at the current iterate, fits a single scalar s_k to set its amplitude, and freezes the update $U_k = s_k\hat{u}_k\hat{v}_k^\top$ before moving on. The procedure is appealing for memory-bounded fine-tuning, for tasks whose target rank is unknown, and as a nonlinear counterpart to classical Hotelling deflation. For a single-layer Gaussian Leaky-ReLU teacher–student model with rank-$r^\star$ target and slope $\alpha \in [0, 1)$, we prove that the rescaled negative *population* gradient equals the current residual plus a nonlinearity-induced bias controlled by a sector angle; the empirical gradient on a fresh batch concentrates around this population surrogate at rate $\gamma_N = \widetilde{O}(\sqrt{d/N})$. This yields a per-stage recovery theorem under a singular-value gap and an additive cumulative bound across r stages, with the final parameter error bounded by the oracle rank-r tail plus accumulated local errors. Sharpness examples show that gap-scale perturbations can flip the top dyad in the main surrogate problem; a separate frozen-feature example shows that covariance control is necessary for the bridge. The theorem covers a *direction-locked* variant in which only the scalar amplitude is optimized per stage; free two-factor optimization is accounted for as an additional measured rotation term that the theorem does not itself bound. This is not a convergence theorem for LoRA in deep networks; the formal contribution is a local residual-surrogate theorem for a direction-locked rank-one procedure in a single-layer Gaussian Leaky-ReLU teacher–student model. Synthetic experiments are consistent with the per-stage diagnostic, while frozen-feature image experiments illustrate a secondary active-state bridge.

1 Introduction

Sequential rank-one learning is the family of methods that build a low-rank object one rank-one piece at a time, each piece chosen to account for what the earlier pieces left behind. It is appealing for three reasons that recur across applications. First, a sequential procedure produces an intermediate-rank frontier along the same training path, without retraining separate models from scratch; this gives a one-run rank sweep, although we do not provide a practical stopping rule for unknown target rank. Second, only one rank-one piece is being optimized at any moment, so the active optimizer-state memory for the current stage stays bounded by a single rank-one factor pair regardless of the final rank. This is an active-state advantage only: total memory, gradient computation, top-dyad extraction, and storage of the frozen accumulated rank-one updates from earlier stages are not reduced. Third,

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the analysis tradition for such procedures is mature [Liao et al., 2024, 2025, Alizadeh Vandchali et al., 2025]: errors made at one stage feed into the residual the next stage operates on, which makes them a natural fit for deflation-style perturbation analysis going back to Hotelling [1933].

Low-Rank Adaptation (LoRA, Hu et al., 2022) reparameterizes a fine-tuned weight as $\mathbf{W}_0 + BA^\top$ with a frozen base $\mathbf{W}_0 \in \mathbb{R}^{d\times m}$ and trainable factors $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{m\times r}$, so the adapter $\Delta = BA^\top$ has rank at most r. Standard LoRA and its variants [Dettmers et al., 2023, Zhang et al., 2023, Valipour et al., 2023, Ding et al., 2023, Benedek and Wolf, 2024, Liu et al., 2024, Meng et al., 2024, Wang et al., 2024, Zhao et al., 2024, He et al., 2025, Zhang et al., 2025] train the rank-r factorization jointly. We study a single-layer model in which the same recipe is run sequentially: one rank-one piece at a time. Beyond the generic motivations above, the LoRA-specific consideration that distinguishes the sequential variant is that the joint factorization $\Delta = BA^\top$ has a rotational ambiguity: any orthogonal $Q \in \mathbb{R}^{r\times r}$ takes (B, A) to (BQ, AQ) without changing Δ, whereas fixing each rank-one direction at the time it is added selects a specific representative of the factorization.

Problem statement. We work with a single weight matrix passed through Leaky ReLU. The teacher is $\widetilde{y} = \sigma_\alpha(\mathbf{x}^\top(\mathbf{W}_0 + \Delta))$ on Gaussian inputs $\mathbf{x} \sim \mathcal{N}(0, I_d)$, where $\sigma_\alpha(z) = \max\{z, \alpha z\}$ is Leaky ReLU with slope $\alpha \in [0, 1)$, the base weight $\mathbf{W}_0$ is frozen, and the target adapter $\Delta \in \mathbb{R}^{d\times m}$ has rank $r^\star$. The *adaptation loss* is the empirical squared-error mismatch between teacher and student outputs,

$$\mathcal{L}_N(\mathbf{W}) = \frac{1}{2N}\sum_{i=1}^{N} \left\|\sigma_\alpha(\mathbf{x}_i^\top \mathbf{W}) - \sigma_\alpha(\mathbf{x}_i^\top(\mathbf{W}_0 + \Delta))\right\|_2^2 .$$

A sequential rank-one algorithm builds the learned adapter $\widehat{\Delta}^{(r)}$ as a sum $\widehat{\Delta}^{(r)} = \sum_{k=1}^{r} U_k$ of r rank-one updates $U_k = s_k \hat{u}_k \hat{v}_k^\top$, one per stage. At stage k it performs three operations. (i) It looks at the gradient of $\mathcal{L}_N$ at the current iterate $\mathbf{W}_0 + \widehat{\Delta}^{(k-1)}$; we call the matrix $\Delta - \widehat{\Delta}^{(k-1)}$ the *residual*, since it is what the adapter still has to recover. The analysis additionally tracks an *oracle residual* $\Delta_k^\star$ along an idealized path that subtracts the exact best rank-one approximation at each stage; the actual and oracle residuals coincide when the algorithm's updates match the oracle. (ii) It picks a unit-norm pair $(\hat{u}_k, \hat{v}_k)$ from this loss gradient: concretely, the top left and right singular vectors of a rescaled gradient, with $\hat{u}_k \in \mathbb{R}^d$ left and $\hat{v}_k \in \mathbb{R}^m$ right. (iii) It fits the per-stage scale s_k, a single scalar, to set the magnitude of the new rank-one update $U_k = s_k \hat{u}_k \hat{v}_k^\top$. The update is then frozen and the algorithm proceeds to stage $k{+}1$ on the new residual. After r stages the learned adapter has the same $BA^\top$ structure as joint LoRA but is built one rank-one piece at a time. We analyze the *direction-locked* variant of this procedure: once the rank-one direction $(\hat{u}_k, \hat{v}_k)$ is read off the gradient, only the scalar amplitude s_k is optimized; the direction itself is not refit. Free two-factor optimization, which can rotate away from the gradient dyad, is not covered by the per-stage theorem; Corollary A.1 provides only a deterministic diagnostic decomposition with a measured rotation term, and the paper does not prove that this rotation term is small.

What changes with a nonlinearity. The linear special case (replace σ_α by the identity) has just been resolved by Alizadeh Vandchali et al. [2025], who give matching error-propagation bounds for sequential rank-one regression. With a nonlinearity in place, three concrete things change. First, the algorithm no longer sees the residual directly: it has to infer $\Delta - \widehat{\Delta}^{(k-1)}$ from the gradient of $\mathcal{L}_N$. In the linear population case the gradient is a clean linear image of the residual; in the nonlinear case the gradient is a linear image of the residual plus an additional bias, as the next point makes precise. Second, the rescaled negative *population* gradient $-\nabla\mathcal{L}_{\text{pop}}/\mu_\alpha$ (with $\mu_\alpha = (1 + \alpha^2)/2$) equals the residual plus an activation-induced bias whose size is controlled by a sector angle between teacher and student columns; the empirical gradient on a fresh batch differs from this population surrogate by sampling error. Reading a rank-one direction off the empirical gradient therefore reads the residual only up to that bias and the sampling error. Third, the residual update is still algebraically additive, $R_{k+1}^W = R_k^W - U_k$, but the algorithm no longer observes the residual: it observes a biased, noisy gradient surrogate. Stage errors therefore propagate through both the residual recursion and the nonlinearity-and-sampling map from residual to surrogate, and the linear-case error-propagation argument of Alizadeh Vandchali et al. [2025] has to be reopened.

The sequential rank-one idea is classical. Hotelling deflation [Hotelling, 1933] is the original procedure: extract the top eigenpair, subtract its outer product from the matrix, and recurse. Mackey [2008] extended deflation to sparse PCA, where the inexactness of each stage forced a careful look

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at how errors compound. Subsequent work has sharpened that picture: Liao et al. [2024] analyze error propagation for inexact Hotelling deflation in PCA, and Alizadeh Vandchali et al. [2025] carry the same analysis through to sequential rank-one regression in the linear setting. A parallel line studies multi-component versions: Gemp et al. [2021, 2022] cast PCA as a multi-player game, and Liao et al. [2025] give the first provable parallel deflation algorithm for distributed PCA. Within parameter-efficient fine-tuning, ReLoRA [Lialin et al., 2023] restarts low-rank updates and merges them, which is sequential rank-r rather than sequential rank-one but shares the philosophy. Our work picks up exactly where Alizadeh Vandchali et al. [2025] stop: their analysis is for linear regression; we ask the same questions when a Leaky-ReLU sits between the input and the loss.

Contributions.

- A population-gradient identity (Lemma 4.3) showing that the rescaled *negative* population gradient $-\nabla\mathcal{L}_{\mathrm{pop}}/\mu_\alpha$ at any iterate equals the current residual plus a controlled remainder driven by a sector angle, together with a finite-sample concentration bound (Lemma 4.4) at rate $\gamma_N = O(\sqrt{d\log(dm(T+1)/\delta)/N})$ for the empirical gradient on a fresh batch.
- A per-stage recovery theorem (Proposition 4.5): when the oracle residual has a singular-value gap and the gradient surrogate remains close to that oracle residual, the rank-one update U_k that the algorithm freezes is close to the next oracle rank-one component, with the error governed by the remainder of Lemma 4.3.
- An additive error-propagation theorem (Proposition 4.6) showing that per-stage errors compound additively in Frobenius norm across r stages, together with a conservative all-stage certificate (Theorem A.4) with explicit per-stage envelope d_k and a stage-wise budget (Corollary A.6). These use oracle quantities $\mathcal{T}_k^\star, \kappa_k^\star, \rho_k^\star$ and the column-norm constant $m_{\min}$, and are post-hoc analytical tools rather than observable stopping rules.
- Sharpness examples (Proposition 4.2): gap-scale perturbations can flip the top dyad in the main surrogate problem, and a separate frozen-feature example shows that covariance control is necessary for the bridge; neither condition can be removed in worst case.
- Synthetic experiments give diagnostic evidence for the per-stage mechanism: a separated rank-3 calibration illustrates the certified regime, and the harder rank-8 run is reported as diagnostic evidence outside the conservative sufficient regime. Frozen-feature experiments illustrate the active-state/rank-frontier tradeoff.

2 Related Work

Greedy low-rank and atomic methods. The closest mathematical neighbors are methods that select rank-one directions from a residual or from first-order information: deflation [Hotelling, 1933, Mackey, 2008], matching pursuit [Mallat and Zhang, 1993], Frank-Wolfe and atomic decomposition [Jaggi and Sulovský, 2010, Jaggi, 2013, Shalev-Shwartz et al., 2011, Lee and Bresler, 2009], and low-rank optimization over convex surrogates [Journée et al., 2010], with related analyses of LoRA expressivity and matrix-factorization dynamics [Zeng and Lee, 2024, Li et al., 2021]. These works typically study a linear inverse problem, a convex matrix objective, or a directly observed residual. Our setting is different: the residual is hidden, and inferring it through a nonlinear teacher-student gradient brings in sector angle, sampling noise, activation bias, and spectral gaps as additional ingredients in the rank-one recovery guarantee.

Low-rank fine-tuning and gradient/spectral subspace selection. A related primitive appears in low-rank fine-tuning: choose, grow, prune, initialize, or project low-rank subspaces using gradient or spectral structure. LoRA [Hu et al., 2022] and related PEFT methods [Houlsby et al., 2019, Lester et al., 2021, Li and Liang, 2021, Ben Zaken et al., 2022, Dettmers et al., 2023] adapt a frozen weight through low-rank, prompt, prefix, quantization, or bias updates; rank-adaptive variants [Valipour et al., 2023, Zhang et al., 2023, Ding et al., 2023, Benedek and Wolf, 2024, Liu et al., 2024, Lialin et al., 2023] grow, prune, or freeze low-rank capacity online; and gradient- or spectrum-driven initializations [Meng et al., 2024, Wang et al., 2024, Zhao et al., 2024, He et al., 2025] use first-order or spectral structure to pick or project the low-rank subspace. These works use gradients and spectra as algorithmic objects in deep models. We study a narrower teacher-student setting that isolates the local question of when a gradient actually reveals the next rank-one residual direction.

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Algorithm 1 Fresh-batch direction-locked sequential rank-one recovery

Require: base weight $\mathbf{W}_0$, target rank r, independent stage batches $\mathcal{B}_1, \dots, \mathcal{B}_r$, scalar-solver steps $t_1, \dots, t_r$

$\mathbf{W}^{\langle 1 \rangle} \leftarrow \mathbf{W}_0$
for $k = 1, \dots, r$ **do**
  $G_k \leftarrow \nabla \mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k \rangle})$ and $A_k \leftarrow -G_k/\mu_\alpha$, where $\mu_\alpha = (1+\alpha^2)/2$
  $(\hat{u}_k, \hat{v}_k) \leftarrow$ top left/right singular vectors of A_k
  Initialize $s_k \leftarrow \sigma_1(A_k)$
  Refine s_k for t_k scalar steps on $\mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k \rangle} + s_k \hat{u}_k \hat{v}_k^\top)$ ($t_k = 0$ keeps the SVD-derived s_k)
  $U_k \leftarrow s_k \hat{u}_k \hat{v}_k^\top$ and $\mathbf{W}^{\langle k+1 \rangle} \leftarrow \mathbf{W}^{\langle k \rangle} + U_k$
end for
return $\widehat{\Delta}^{(r)} = \sum_{k=1}^{r} U_k$

3 Setup and algorithm

Let $\sigma_\alpha(z) = \max\{z, \alpha z\}$ with $\alpha \in [0, 1)$, applied coordinatewise. Inputs satisfy $\mathbf{x} \sim \mathcal{N}(0, I_d)$. The teacher is

$$\widetilde{y} = \sigma_\alpha(\mathbf{x}^\top \widetilde{\mathbf{W}}^\star), \qquad \widetilde{\mathbf{W}}^\star = \mathbf{W}_0 + \Delta, \qquad \Delta \in \mathbb{R}^{d \times m}, \quad \operatorname{rank}(\Delta) = r^\star.$$

Given N samples $(\mathbf{x}_i, \widetilde{y}_i)$, define

$$\mathcal{L}_N(\mathbf{W}) = \frac{1}{2N} \sum_{i=1}^{N} \left\| \sigma_\alpha(\mathbf{x}_i^\top \mathbf{W}) - \sigma_\alpha(\mathbf{x}_i^\top \widetilde{\mathbf{W}}^\star) \right\|_2^2.$$

The population loss is denoted $\mathcal{L}_{\mathrm{pop}}(\mathbf{W}) = \mathbb{E}[\mathcal{L}_N(\mathbf{W})]$. At the kink of σ_α, empirical gradients use an arbitrary fixed subgradient value in $[\alpha, 1]$; all concentration statements below use the same convention. Away from the kink this is the ordinary gradient.

The sequential learner writes $\widehat{\Delta}^{(r)} = \sum_{k=1}^{r} U_k$, where each $U_k = b_k a_k^\top$ has rank at most one. When the rank-one direction is locked to the gradient's top dyad, we write $U_k = s_k \hat{u}_k \hat{v}_k^\top$ with unit-norm $\hat{u}_k \in \mathbb{R}^d$, $\hat{v}_k \in \mathbb{R}^m$ and a single scalar amplitude s_k; this is the form analyzed by Algorithm 1 and Proposition 4.5. The general $b_k a_k^\top$ form is used in Corollary A.1, which gives only a deterministic diagnostic decomposition with a measured rotation term for free-factor solvers that may rotate away from the gradient dyad. Let

$$\mathbf{W}^{\langle k \rangle} = \mathbf{W}_0 + \sum_{j<k} U_j, \qquad R_k^W = \Delta - \sum_{j<k} U_j.$$

The oracle residual is $\Delta_1^\star = \Delta$ and $\Delta_{k+1}^\star = \Delta_k^\star - U_k^\star$, where $U_k^\star$ is the best rank-one approximation of $\Delta_k^\star$. Write

$$\mathcal{T}_k^\star = \min\{\sigma_1(\Delta_k^\star) - \sigma_2(\Delta_k^\star), \sigma_1(\Delta_k^\star)\}.$$

Singular values beyond rank are zero. If $\Delta_k^\star = 0$, then $\mathcal{T}_k^\star = 0$ and the perturbation results do not apply.

The scalar budgets t_k are optimization hyperparameters, not part of the theory. Appendix B gives the scalar-solver details used for the ablation figures. If $r > r^\star$ (over-specified rank), the algorithm continues for all r stages, but the perturbation results in Section 4 apply only at stages $k \le r^\star$ where $\mathcal{T}_k^\star > 0$.

4 Theory

The theory separates the perturbation principle from the model-specific gradient calculation. First, any matrix surrogate close to the oracle residual reveals the next singular dyad. The Gaussian Leaky-ReLU model then supplies one concrete surrogate: a scalar-rescaled fresh-batch gradient. A smooth matched model supplies another through Hessian preconditioning. The final part of the section closes these local statements into a conservative multi-stage certificate.

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4.1 Rank-one recovery from a residual surrogate

Theorem 4.1 (Surrogate-to-dyad principle)**.** *Let A_k be any matrix and assume $\mathcal{T}_k^\star > 0$. If*

$$\|A_k - \Delta_k^\star\|_{\mathrm{F}} \le \varepsilon_k \le \frac{1}{4}\mathcal{T}_k^\star,$$

then the top singular dyad $\hat{u}_k\hat{v}_k^\top$ of A_k obeys

$$\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \le C\frac{\varepsilon_k}{\mathcal{T}_k^\star}.$$

If additionally $U_k = s_k\hat{u}_k\hat{v}_k^\top$ with $|s_k - \sigma_1(A_k)| \le \varepsilon_{k,\mathrm{amp}}$, then

$$\|U_k - U_k^\star\|_{\mathrm{F}} \le \varepsilon_{k,\mathrm{amp}} + C'\frac{\sigma_1(\Delta_k^\star)}{\mathcal{T}_k^\star}\varepsilon_k.$$

This theorem is the abstract perturbation step behind the paper. The remaining results identify settings in which a gradient-based surrogate A_k is close to $\Delta_k^\star$. The following proposition records the matching sharpness: the gap dependence in Theorem 4.1 cannot be removed in worst case, and a companion construction shows the analogous covariance condition for the frozen-feature bridge of Appendix D.

Proposition 4.2 (Sharpness of gap and covariance requirements)**.** *The gap and covariance terms in the surrogate and bridge bounds cannot be removed in worst case.*

1. *Let $\Delta = \sigma_1 u_1 v_1^\top + \sigma_2 u_2 v_2^\top$, where u_1, u_2 and v_1, v_2 are orthonormal, $\sigma_1 > \sigma_2 \ge 0$, and $g = \sigma_1 - \sigma_2$. For any $0 < \varepsilon < (\sigma_1 + \sigma_2)/2$, there exists E with $\|E\|_2 = g/2 + \varepsilon$ and $\|E\|_{\mathrm{F}} = \sqrt{2}(g/2 + \varepsilon)$ such that the top singular dyad of $\Delta + E$ is $u_2 v_2^\top$, not the oracle dyad $u_1 v_1^\top$.*
2. *More generally, in the deterministic perturbation model $A = \Delta + E$, $\|E\|_2 \le \varepsilon$, no estimator of the top dyad from A can improve the gap scaling uniformly over rank-two residuals with gap g: for a universal $c > 0$,*

$$\inf_{\Phi}\sup_{\Delta,E}\left\|\Phi(\Delta + E) - u_1 v_1^\top\right\|_{\mathrm{F}} \ge c\min\{1, \varepsilon/g\},$$

   where the supremum ranges over rank-two Δ with $\sigma_1(\Delta) - \sigma_2(\Delta) = g$ and all perturbations $\|E\|_2 \le \varepsilon$.
3. *Consider the frozen-feature squared-loss bridge of Appendix D, in which a linear head is fit on frozen features with covariance $\Sigma = \mathbb{E}[zz^\top]$ and the population negative gradient at residual R is ΣR. Take $R = e_1 e_1^\top + \rho e_2 e_2^\top$ with $0 < \rho < 1$ and $\Sigma = \mathrm{diag}(1, \lambda)$ with $\lambda > 1/\rho$. Then the residual top dyad is $e_1 e_1^\top$, but the population negative gradient ΣR has top dyad $e_2 e_2^\top$. Moreover $\|\Sigma - I\|_2 = \lambda - 1$ can be arbitrarily close to $(1-\rho)/\rho$, which is of the same order as the residual gap when ρ is close to one.*

Thus top-dyad recovery cannot be certified from perturbation size without a gap-scale condition, and the frozen-feature bridge genuinely requires covariance control.

4.2 Gaussian Leaky-ReLU scalar-gradient surrogate

Lemma 4.3 (Population gradient decomposition)**.** *For the Leaky-ReLU teacher-student loss, suppose every nonzero teacher column has a nonzero learner column: $w_j \ne 0$ whenever $\widetilde{w}_j^\star \ne 0$. Then*

$$\nabla\mathcal{L}_{\mathrm{pop}}(\mathbf{W}) = \mu_\alpha(\mathbf{W} - \widetilde{\mathbf{W}}^\star) + \Psi_\alpha(\mathbf{W}), \qquad \mu_\alpha = \frac{1+\alpha^2}{2}.$$

If, in addition, every nonzero teacher column satisfies $\angle(\mathbf{w}_j, \widetilde{w}_j^\star) \le \theta_0 < \pi/2$, then

$$\|\Psi_\alpha(\mathbf{W})\|_{\mathrm{F}} \le (1-\alpha)^2 C_0\theta_0\left\|\mathbf{W} - \widetilde{\mathbf{W}}^\star\right\|_{\mathrm{F}}, \qquad C_0 = \frac{1}{2\pi}\sqrt{1 + \frac{4}{\pi^2}}.$$

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Lemma 4.4 (Gradient concentration on a fixed or fresh-batch path). *Fix a path $\{\mathbf{W}_t\}_{t=0}^{T} \subset \mathbb{R}^{d\times m}$ independent of the sample used to estimate the gradients. In particular, the statement applies conditionally at a stage when a fresh batch is drawn after the previous stages have fixed the current iterate. In this lemma, $\nabla\mathcal{L}_N$ denotes the selected empirical gradient obtained by the fixed kink convention above, and $\nabla\mathcal{L}_{\mathrm{pop}}$ denotes the expectation of the same selected gradient; at differentiability points the selected and ordinary population gradients coincide. There are universal constants C_1, C_2 such that if*

$$N \geq C_1 d \log\left(\frac{10dm(T+1)}{\delta}\right),$$

then with probability at least $1-\delta$,

$$\max_{0\leq t\leq T} \|\nabla\mathcal{L}_N(\mathbf{W}_t) - \nabla\mathcal{L}_{\mathrm{pop}}(\mathbf{W}_t)\|_{\mathrm{F}} \leq \gamma_N \max_{0\leq t\leq T} \left\|\mathbf{W}_t - \widetilde{\mathbf{W}}^\star\right\|_{\mathrm{F}},$$

where

$$\gamma_N = C_2\sqrt{\frac{d\log\left(\frac{10dm(T+1)}{\delta}\right)}{N}}.$$

Proposition 4.5 (One-stage perturbation bound). *Assume $\mathcal{T}_k^\star > 0$, assume the sector condition of Lemma 4.3 holds at $\mathbf{W}^{\langle k\rangle}$, and assume the batch used for the stage-k gradient is independent of $\mathbf{W}^{\langle k\rangle}$ as in Lemma 4.4. Work on the concentration event of Lemma 4.4 and set*

$$\eta_\alpha(\theta_0, \gamma_N) = \frac{(1-\alpha)^2 C_0\theta_0 + \gamma_N}{\mu_\alpha}.$$

Let $A_k = -\nabla\mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k\rangle})/\mu_\alpha$. If

$$\eta_\alpha \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}} \leq \frac{\mathcal{T}_k^\star}{4},$$

then the top singular dyad $\hat{u}_k\hat{v}_k^\top$ of A_k obeys

$$\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \leq C\frac{\eta_\alpha \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}}{\mathcal{T}_k^\star},$$

where $u_{1,k}, v_{1,k}$ are the leading singular vectors of $\Delta_k^\star$. Moreover, if $U_k = s_k\hat{u}_k\hat{v}_k^\top$ and $\varepsilon_{k,\mathrm{amp}} = |s_k - \sigma_1(A_k)|$, then

$$\|U_k - U_k^\star\|_{\mathrm{F}} \leq \varepsilon_{k,\mathrm{amp}} + C'\frac{\eta_\alpha \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}}{\mathcal{T}_k^\star}\sigma_1(\Delta_k^\star).$$

Locality and the sector condition. Proposition 4.5 is a *local* stage certificate: Algorithm 1 does not enforce the sector condition, and the proposition only certifies a stage when $\mathbf{W}^{\langle k\rangle}$ stays in a small-sector neighborhood of the teacher and the surrogate is accurate relative to $\mathcal{T}_k^\star$.

Reading the sufficient condition. The condition $\eta_\alpha \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}} \leq \mathcal{T}_k^\star/4$ has two pieces. The first, $\eta_\alpha \left\|R_k^W\right\|_{\mathrm{F}}$, is the nonlinearity-plus-sampling slack scaled by the current residual norm; it shrinks as $\alpha \to 1$ (less ReLU bias) and as N grows (smaller γ_N). The second, $\left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}$, is the accumulated drift between the actual residual and the oracle path: it is zero at stage 1 (where $R_1^W = \Delta = \Delta_1^\star$) and grows from earlier stages when the algorithm's updates differ from $U_j^\star$. Both pieces must remain a constant fraction of the current oracle gap $\mathcal{T}_k^\star$.

Stage-wise sector check. The sector assumption in Proposition 4.5 is automatic if $m_{\min} = \min_{j:\widetilde{w}_j^\star\neq 0}\left\|\widetilde{w}_j^\star\right\|_2 > 0$ and $\left\|R_k^W\right\|_{\mathrm{F}} = \left\|\widetilde{W}^\star - \mathbf{W}^{\langle k\rangle}\right\|_{\mathrm{F}} \leq m_{\min}\sin\theta_0$; Appendix C.1 gives the one-line proof and the corresponding all-stage envelope check.

Proposition 4.5 analyzes the direction-locked update $U_k = s_k\hat{u}_k\hat{v}_k^\top$ produced by Algorithm 1. A stage optimizer that freely updates both factors b_k, a_k can rotate away from the gradient dyad and is not covered; Corollary A.1 (App. A) records a deterministic diagnostic decomposition with a measured rotation term and does not prove that this rotation term is small. The same surrogate-to-dyad principle also applies to smooth matched activations through oracle Hessian preconditioning (Proposition A.2, App. A), which scopes the mechanism beyond Leaky-ReLU.

Proofs of Theorem 4.1 and Proposition A.2 are in Appendix A.

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4.3 Multi-stage propagation and the all-stage certificate

Proposition 4.6 (Additive end-to-end propagation)**.** *For any sequence of rank-one increments* $U_1, \dots, U_r$,

$$\left\|\Delta - \widehat{\Delta}^{(r)}\right\|_{\mathrm{F}} \le \left(\sum_{j>r} \sigma_j(\Delta)^2\right)^{1/2} + \sum_{k=1}^{r} \|U_k - U_k^\star\|_{\mathrm{F}} .$$

Consequently, if the condition in Proposition 4.5 holds for all k*, the final error is the oracle rank-*r *tail plus the sum of the stage-wise perturbation terms.*

Proposition 4.7 (Prediction loss from parameter error)**.** *For any* $\mathbf{W}$ *and any* $\alpha \in [0, 1]$,

$$\mathcal{L}_{\mathrm{pop}}(\mathbf{W}) \le \frac{1}{2} \left\|\mathbf{W} - \widetilde{\mathbf{W}}^\star\right\|_{\mathrm{F}}^2 .$$

Consequently, the parameter-error bound in Proposition 4.6 implies the corresponding population-prediction bound obtained by squaring its right hand side and multiplying by $1/2$.

Iterating Proposition 4.5 along the additive bound of Proposition 4.6 yields a closed-form end-to-end certificate for the Gaussian Leaky-ReLU instantiation. Define $d_1 = 0$ and the recurrence

$$d_{k+1} = \big(1 + C'\rho_k^\star(1+\eta_k)\big)d_k + \varepsilon_{k,\mathrm{amp}} + C'\eta_k\kappa_k^\star s_k^\star,$$

with $\rho_k^\star = s_k^\star/\mathcal{T}_k^\star$ and $\kappa_k^\star = \|\Delta_k^\star\|_{\mathrm{F}}/\mathcal{T}_k^\star$. Whenever the per-stage feasibility condition $\eta_k(\kappa_k^\star + d_k/\mathcal{T}_k^\star) + d_k/\mathcal{T}_k^\star \le 1/4$ holds at every stage, d_{r+1} bounds the cumulative stage-error sum in Proposition 4.6, and the final parameter error is the oracle rank-r tail plus d_{r+1}. The full theorem (Theorem A.4, App. A) makes the recurrence explicit; Corollary A.5 reads it as a uniform fresh-batch sample bound at rate $\gamma = \widetilde{O}(\sqrt{d/N})$, and Corollary A.6 converts it into a stage-wise margin $M_k > 0$ that certifies an additional direction at stage k from oracle quantities. This certificate should be read as a conservative sufficient condition rather than a generic convergence theorem: past mismatch enters in Frobenius norm, so a stage that consumes a sizeable fraction of its gap budget tightens every later stage's margin, and the algorithm does not enforce sector control along the path. A companion corollary (Corollary A.3, App. A) records the population-prediction bound for a fixed-direction scalar refit, where only ε-optimality of the scalars is assumed.

The amplification factor $1 + C'\rho_k^\star(1+\eta_k)$ compounds, so Theorem A.4 is informative only when $\rho_k^\star = O(1)$, $\eta_k \ll 1$, and d_k stays below $\mathcal{T}_k^\star$ throughout; outside this regime the bound can become vacuous from sample size alone, and Corollary A.6 should be read as an oracle stage-by-stage analytical tool in $\mathcal{T}_k^\star, \kappa_k^\star, \rho_k^\star, m_{\min}$, not an observable stopping rule.

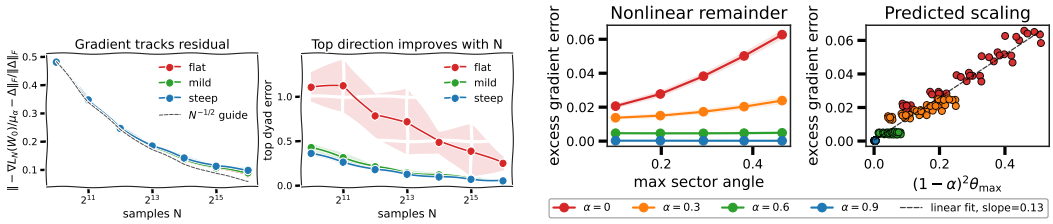
5 Experiments

We organize the synthetic experiments around the structure of the proof. We first open the loop at stage 1 and ask whether the rescaled gradient behaves as the residual surrogate of Lemmas 4.3–4.4. We then run Algorithm 1 on a rank-8 teacher and check whether the per-stage sufficient condition of Proposition 4.5 holds at every stage. Finally we isolate the residual spectral gap and check that the top-dyad error scales the way Theorem 4.1 predicts. Appendix B extends the same story with calibration, sample-scaling, rank-mismatch, activation-slope, scalar-solver, and robustness diagnostics. Appendix D reports a frozen-feature bridge whose Gaussian-input assumption no longer holds. We use it to test the bridge propositions of Section D.1 rather than the main theorem.

Stage-1 residual surrogate. At $\mathbf{W}_0$ we compare the rescaled empirical gradient to the true residual as we sweep the fresh batch size N (Figure 1(a)). Both the relative gradient–residual error and the top-dyad error decrease with N at the $\sqrt{d/N}$ rate of Lemma 4.4, consistent with the stage-1 conclusion of Theorem 4.1. In Figure 1(b) we probe the proportional dependence in the remainder bound of Lemma 4.3. After we subtract the α=1 sampling baseline, the excess gradient–residual error scales linearly with $(1-\alpha)^2\theta_{\max}$. We use an excess-error proxy and therefore test the predicted scaling, not the literal constant C_0.

Diagnostics along the path on a harder rank-8 run. We then run the full sequential algorithm on a deliberately harder rank-8 teacher and track the empirical perturbation ratio, the maximum sector angle, and the scalar slack $\varepsilon_{k,\mathrm{amp}}$ at every stage (Figure 2). Rank-8 is a stress test: residual gaps are

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(a) Gradient as residual.

(b) Nonlinear remainder.

Figure 1: Mechanism checks at $\mathbf{W}_0$ for steep/mild/flat teacher spectra (large/moderate/small singular-value gaps). **(a) Gradient as residual.** Left: relative gradient–residual error $\|-\nabla\mathcal{L}_N(\mathbf{W}_0)/\mu_\alpha - \Delta\|_{\mathrm{F}}/\|\Delta\|_{\mathrm{F}}$ decreases with the fresh batch size N; the dashed curve is an $N^{-1/2}$ rate guide for the sampling term in Lemma 4.4, not a finite-sample upper bound. Right: top singular dyad error at the same iterate decreases with N, consistent with the stage-1 conclusion of Theorem 4.1 composed with Proposition 4.5. **(b) Nonlinear remainder.** Left: after subtracting the $\alpha{=}1$ sampling baseline, the excess relative gradient–residual error grows with the maximum sector angle for fixed α, with the gap closing as $\alpha \to 1$. Right: pooled scatter against $(1-\alpha)^2\theta_{\max}$ across $(\alpha, \|\Delta\|_{\mathrm{F}})$ settings, with linear fit (slope 0.13). The figure tests the proportional dependence in the remainder bound of Lemma 4.3, $\|\Psi_\alpha\|_{\mathrm{F}} \propto (1-\alpha)^2\theta_0\|\mathbf{W} - \widetilde{\mathbf{W}}^\star\|_{\mathrm{F}}$, using an excess-error proxy; it is consistent with the predicted scaling but does not certify the literal constant C_0.

smaller, so the per-stage condition is harder to satisfy. The empirical ratio sits above the sufficient $1/4$ threshold of Proposition 4.5 for most of the run on the mild and flat spectra and approaches the threshold only in later stages on steep. Stages above $1/4$ are outside the conservative sufficient regime and are not certified by Proposition 4.5; this does not contradict the theory because the proposition is sufficient, not necessary, and we report this figure as diagnostic evidence. The cleanly certified regime, with ratio below $1/4$ at every stage, is the separated rank-3 calibration with large N in Figure 3 (App. B).

Gap is the dominant lever. Theorem 4.1 predicts that the top-dyad error scales as the gap-normalized perturbation ratio, and Proposition 4.2 shows that the gap requirement cannot be removed in worst case. Figure 4 (App. B) tests this prediction with a controlled sweep that varies only the residual spectral gap and finds the top-dyad error grows roughly linearly with the perturbation ratio.

End-to-end sample dependence closes the chain. The previous three panels test a single link of the proof at a time. We close the chain in Figure 6 by sweeping the fresh stage-batch size N on the rank-8 run. Larger N tightens the maximum-over-stages perturbation ratio (right), as Lemma 4.4 predicts, and improves both final parameter recovery (left) and held-out prediction loss (middle). We compare the observed log-log slope to the $N^{-1/2}$ guide of Lemma 4.4. Agreement is qualitative because final error also includes nonlinear bias, gap effects, scalar slack, and accumulated past-stage mismatch. Read together, the four panels move from the gradient-as-residual identity (Lemma 4.3, Lemma 4.4) through stage-wise dyad recovery (Proposition 4.5) to additive end-to-end propagation (Proposition 4.6, Proposition 4.7).

6 Discussion

The main message is theoretical: in nonlinear low-rank models, a gradient reveals the next rank-one residual direction only when the chosen gradient surrogate is accurate relative to the residual spectral gap. Sector control keeps the gradient identity tight, fresh batches keep concentration unconditional, and gap separation keeps the SVD perturbation argument quantitative. Proposition 4.2 shows that the gap requirement and the frozen-feature covariance requirement cannot be removed in worst case.

Implication for the LoRA initialization step. The formal contribution analyzes a local question shared with gradient- and spectrum-based LoRA initialization schemes (PiSSA, LoRA-GA, LoRA-One, GaLore): when does the top dyad of the current loss gradient approximate the next residual direction? Proposition 4.5 gives a sufficient local condition for that selection step in our single-layer Gaussian Leaky-ReLU model. Free-factor optimization that follows initialization is post-init behavior

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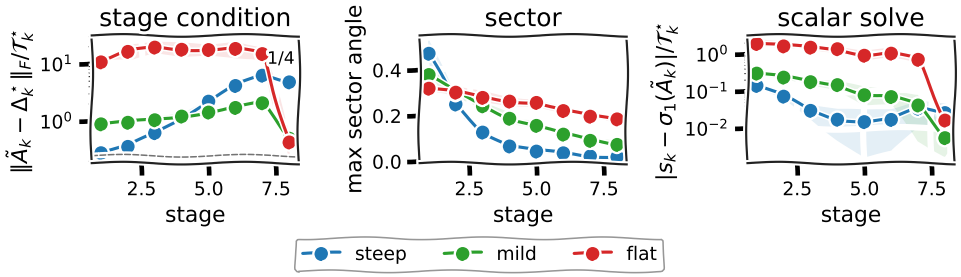


Figure 2: Per-stage diagnostics for the analyzed direction-locked algorithm on a harder rank-8 run, steep/mild/flat teacher spectra (large suite). Left: empirical perturbation ratio $\|\widetilde{A}_k - \Delta_k^\star\|_{\mathrm{F}}/\mathcal{T}_k^\star$ at each stage; the dashed line marks the $1/4$ sufficient threshold of Proposition 4.5. Middle: maximum sector angle along teacher columns, the quantity that controls the remainder bound in Lemma 4.3. Right: scalar-amplitude mismatch $|s_k - \sigma_1(\widetilde{A}_k)|$, the per-stage slack $\varepsilon_{k,\mathrm{amp}}$ that enters Proposition 4.5 additively. Rank-8 is a diagnostic stress test: the mild and flat spectra spend most of the run with perturbation ratio above $1/4$, and the steep spectrum approaches the threshold only in later stages. Stages with ratio above $1/4$ are outside the conservative sufficient regime of Proposition 4.5; this does not contradict the theory because the proposition is sufficient, not necessary. The separately certified regime, where the ratio stays below $1/4$ at every stage, is the rank-3 calibration with large N in Figure 3.

and outside the theorem; the scalar-solver ablations are consistent with the local theory (once the direction is fixed, line search, scalar Adam, and scalar backfitting behave nearly identically and the residual spectral gap dominates the stagewise error) but do not extend the certificate to free two-factor optimization. The frozen-feature experiments are bridge evidence, not a claim that sequential rank-one training dominates simultaneous rank-r optimization.

Open directions. Extending the analysis past the direction-locked constraint, to free two-factor LoRA optimization, requires controlling solver rotation; Corollary A.1 names exactly the term any such extension must bound. Multilayer settings introduce feature coupling and representation drift across stages, neither of which is in scope here; the local mechanism we identify is what such an extension would need to preserve.

7 Limitations

This paper is not a convergence theorem for LoRA in deep networks; the formal contribution is a local residual-surrogate theorem for the gradient-dyad selection step in a direction-locked rank-one procedure on a single-layer Gaussian Leaky-ReLU teacher–student model. The theorems are local, gap-dependent, and model-specific. They assume Gaussian inputs, a single Leaky-ReLU layer, a fresh independent batch per stage, sector control around the teacher, and a separated oracle spectrum at every stage; Lemma C.1 converts an envelope on $\left\|R_k^W\right\|_{\mathrm{F}}$ into the sector condition along the path, but Algorithm 1 does not enforce it. The Gaussian-input assumption is load-bearing for both Lemma 4.3 (via the half-space identity) and Lemma 4.4 (via Gaussian fourth moments); sub-Gaussian inputs would require a different gradient calculation and concentration argument. The per-stage theorem covers the direction-locked update only. Corollary A.1 provides a deterministic diagnostic decomposition for free two-factor optimization with a measured rotation term that we do not prove to be small; free-factor optimization is post-initialization and outside the theorem. The all-stage certificates (Theorem A.4, Corollary A.5, Corollary A.6) use oracle quantities $\mathcal{T}_k^\star, \kappa_k^\star, \rho_k^\star, m_{\min}$ and are post-hoc analytical tools, not observable stopping rules; the envelope is conservative and does not yield a generic convergence theorem from sample size alone. The procedure produces a one-run rank frontier, but we do not provide a practical stopping rule for unknown target rank, and the memory benefit concerns active optimizer-state memory only: total memory, gradient computation, top-dyad extraction, and storage of frozen accumulated updates are not reduced. The smooth extension uses an oracle Hessian preconditioner. We do not prove reused-batch adaptive concentration, multilayer feature coupling, or deep-network LoRA convergence; the frozen-feature image experiments are bridge evidence for the residual-surrogate mechanism, not a deep fine-tuning benchmark.

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A Proofs of Main Theory Results

A.1 Proof of Theorem 4.1

Proof. Weyl's inequality gives

$$|\sigma_i(A_k) - \sigma_i(\Delta_k^\star)| \le \|A_k - \Delta_k^\star\|_2 \le \varepsilon_k.$$

Since $\varepsilon_k \le \mathcal{T}_k^\star/4$, the leading singular subspace remains separated. Wedin's sin-theta theorem applied to the left and right singular subspaces yields

$$\min_{\xi\in\{\pm1\}} \left(\|\hat{u}_k - \xi u_{1,k}\|^2 + \|\hat{v}_k - \xi v_{1,k}\|^2\right)^{1/2} \le C\varepsilon_k/\mathcal{T}_k^\star.$$

Therefore

$$\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \le \|\hat{u}_k - \xi u_{1,k}\| + \|\hat{v}_k - \xi v_{1,k}\| \le C\varepsilon_k/\mathcal{T}_k^\star.$$

For the scaled update,

$$\begin{aligned}\left\|s_k\hat{u}_k\hat{v}_k^\top - \sigma_1(\Delta_k^\star)u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \le |s_k - \sigma_1(\Delta_k^\star)| \\ + \sigma_1(\Delta_k^\star)\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}}.\end{aligned}$$

Another application of Weyl gives $|\sigma_1(A_k) - \sigma_1(\Delta_k^\star)| \le \varepsilon_k$, so $|s_k - \sigma_1(\Delta_k^\star)| \le \varepsilon_{k,\mathrm{amp}} + \varepsilon_k$. Since $\mathcal{T}_k^\star \le \sigma_1(\Delta_k^\star)$, the additive ε_k is absorbed into $C'\sigma_1(\Delta_k^\star)\varepsilon_k/\mathcal{T}_k^\star$. □

A.2 Proof of Proposition 4.2

For the perturbation claim, set $t = g/2 + \varepsilon$ and

$$E = t(u_2v_2^\top - u_1v_1^\top).$$

The two rank-one terms have orthogonal left and right singular directions, so $\|E\|_2 = t$ and $\|E\|_{\mathrm{F}} = \sqrt{2}t$. Moreover

$$\Delta + E = (\sigma_1 - t)u_1v_1^\top + (\sigma_2 + t)u_2v_2^\top.$$

Since $t = g/2 + \varepsilon$,

$$\sigma_2 + t = \frac{\sigma_1 + \sigma_2}{2} + \varepsilon > \frac{\sigma_1 + \sigma_2}{2} - \varepsilon = \sigma_1 - t.$$

The upper bound on ε ensures $\sigma_1 - t > 0$, so the leading singular dyad of $\Delta + E$ is $u_2v_2^\top$. At $\varepsilon = 0$ the two leading singular values tie, so the top dyad is not uniquely determined.

For the minimax perturbation claim, it suffices to work in two dimensions with symmetric positive semidefinite matrices. Let $g > 0$ be fixed and write $q_\theta = (\cos\theta, \sin\theta)$ and $q_\theta^\perp = (-\sin\theta, \cos\theta)$. For any $\theta \in (0, \pi/4]$, define

$$\Delta_\theta = (1+g)q_\theta q_\theta^\top + q_\theta^\perp q_\theta^{\perp\top}, \qquad \Delta_{-\theta} = (1+g)q_{-\theta}q_{-\theta}^\top + q_{-\theta}^\perp q_{-\theta}^{\perp\top}.$$

Both matrices have singular-value gap g, and their top dyads are $D_\theta = q_\theta q_\theta^\top$ and $D_{-\theta} = q_{-\theta}q_{-\theta}^\top$. Since

$$\Delta_\theta - \Delta_{-\theta} = g(D_\theta - D_{-\theta}), \qquad \|D_\theta - D_{-\theta}\|_2 = \sin(2\theta),$$

the common observation $A = (\Delta_\theta + \Delta_{-\theta})/2$ is consistent with both teachers using perturbations of operator norm $g\sin(2\theta)/2$. Choose θ so that $\sin(2\theta) \le \min\{1, 2\varepsilon/g\}$ and this norm is at most ε. For any estimator $\Phi(A)$, the triangle inequality gives

$$\max_{\xi\in\{\theta,-\theta\}} \|\Phi(A) - D_\xi\|_{\mathrm{F}} \ge \frac{1}{2}\|D_\theta - D_{-\theta}\|_{\mathrm{F}} = \frac{\sqrt{2}}{2}\sin(2\theta).$$

Choosing θ with $\sin(2\theta)$ a fixed constant multiple of $\min\{1, \varepsilon/g\}$ proves the displayed lower bound.

For the covariance claim,

$$\Sigma R = e_1e_1^\top + \lambda\rho e_2e_2^\top.$$

The residual R has leading singular value 1 with dyad $e_1e_1^\top$, because $\rho < 1$. If $\lambda > 1/\rho$, then $\lambda\rho > 1$, so the population negative gradient ΣR has leading dyad $e_2e_2^\top$. Finally, $\|\Sigma - I\|_2 = \lambda - 1$, and taking $\lambda \downarrow 1/\rho$ makes this quantity approach $1/\rho - 1 = (1-\rho)/\rho$. If $\rho = 1 - g$, this equals $g/(1-g)$, which is of the same order as the residual gap g for small g.

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A.3 Proof of Lemma 4.3

The loss separates over output coordinates, so it is enough to prove the claim for one column and then sum in Frobenius norm. Fix a teacher column $\mathbf{w}^\star$ and define

$$\ell_\alpha(\mathbf{w}) = \frac{1}{2}\mathbb{E}\big[\big(\sigma_\alpha(\mathbf{x}^\top\mathbf{w}) - \sigma_\alpha(\mathbf{x}^\top\mathbf{w}^\star)\big)^2\big], \qquad h = \mathbf{w} - \mathbf{w}^\star.$$

If $\mathbf{w}^\star = 0$, then

$$\ell_\alpha(\mathbf{w}) = \frac{1}{2}\mathbb{E}[\sigma_\alpha(\mathbf{x}^\top\mathbf{w})^2] = \frac{\mu_\alpha}{2}\|\mathbf{w}\|^2,$$

so $\nabla\ell_\alpha(\mathbf{w}) = \mu_\alpha\mathbf{w} = \mu_\alpha(\mathbf{w} - \mathbf{w}^\star)$ and the remainder is zero. We may therefore assume $\mathbf{w}^\star \neq 0$. The sector condition is not needed for the identity; the nonzero-column hypothesis gives $\mathbf{w} \neq 0$. The sector condition is used only for the final remainder bound. Values of σ'_α at zero are irrelevant for the population identities below; for empirical gradients one may choose any value in $[\alpha, 1]$ at the kink.

Write $\sigma_\alpha(z) = \alpha z + (1-\alpha)\sigma_0(z)$ with $\sigma_0(z) = \max\{z, 0\}$. Expanding the square gives

$$\ell_\alpha(\mathbf{w}) = \frac{\alpha^2}{2}\mathbb{E}[(\mathbf{x}^\top h)^2] + \alpha(1-\alpha)\mathbb{E}[(\mathbf{x}^\top h)(\sigma_0(\mathbf{x}^\top\mathbf{w}) - \sigma_0(\mathbf{x}^\top\mathbf{w}^\star))] + (1-\alpha)^2\ell_0(\mathbf{w}),$$

where

$$\ell_0(\mathbf{w}) = \frac{1}{2}\mathbb{E}[(\sigma_0(\mathbf{x}^\top\mathbf{w}) - \sigma_0(\mathbf{x}^\top\mathbf{w}^\star))^2]$$

is the corresponding ReLU loss. The first term has gradient $\alpha^2 h$. For the mixed term, use the Gaussian half-space identity

$$\mathbb{E}[(\mathbf{x}^\top a)\sigma_0(\mathbf{x}^\top b)] = \frac{1}{2}a^\top b,$$

which follows by conditioning on $\mathbf{x}^\top b$ and using symmetry. Therefore

$$\mathbb{E}[(\mathbf{x}^\top h)(\sigma_0(\mathbf{x}^\top\mathbf{w}) - \sigma_0(\mathbf{x}^\top\mathbf{w}^\star))] = \frac{1}{2}h^\top(\mathbf{w} - \mathbf{w}^\star) = \frac{1}{2}\|h\|^2,$$

and the mixed term contributes $\alpha(1-\alpha)h$ to the gradient.

It remains to record the ReLU gradient exactly. Let $a = \|\mathbf{w}\|$, $b = \|\mathbf{w}^\star\|$, $\theta = \angle(\mathbf{w}, \mathbf{w}^\star)$, $\rho = \cos\theta$, $s = \sin\theta$, $u = \mathbf{w}/a$, and write

$$\mathbf{w}^\star/b = \rho u + sq, \qquad q \perp u, \quad \|q\| = 1.$$

The ReLU kernel for Gaussian inputs is

$$K(\mathbf{w}, \mathbf{w}^\star) = \mathbb{E}[\sigma_0(\mathbf{x}^\top\mathbf{w})\sigma_0(\mathbf{x}^\top\mathbf{w}^\star)] = \frac{ab}{2\pi}\{\sin\theta + (\pi - \theta)\cos\theta\} \quad \text{[Cho and Saul, 2009]}.$$

Since $\mathbb{E}[\sigma_0(\mathbf{x}^\top\mathbf{w})^2] = a^2/2$, we have $\ell_0(\mathbf{w}) = a^2/4 + \text{const} - K(\mathbf{w}, \mathbf{w}^\star)$. Set $f(\theta) = \sin\theta + (\pi - \theta)\cos\theta$. Then $f'(\theta) = -(\pi - \theta)\sin\theta$ and $\nabla_{\mathbf{w}}\theta = -q/a$, so

$$\nabla_{\mathbf{w}}K(\mathbf{w}, \mathbf{w}^\star) = \frac{b}{2\pi}\{f(\theta)u + (\pi - \theta)sq\}.$$

Thus

$$\nabla\ell_0(\mathbf{w}) = \frac{1}{2}\mathbf{w} - \frac{b}{2\pi}\{(\sin\theta + (\pi - \theta)\rho)u + (\pi - \theta)sq\}.$$

Subtracting $\frac{1}{2}h = \frac{1}{2}(a - b\rho)u - \frac{1}{2}bsq$ gives

$$\nabla\ell_0(\mathbf{w}) = \frac{1}{2}h + \Psi_0(\mathbf{w}), \qquad \Psi_0(\mathbf{w}) = \frac{b}{2\pi}\left[(\rho\theta - \sin\theta)u + \theta sq\right].$$

Combining the three contributions,

$$\nabla\ell_\alpha(\mathbf{w}) = \left(\alpha^2 + \alpha(1-\alpha) + \frac{(1-\alpha)^2}{2}\right)h + (1-\alpha)^2\Psi_0(\mathbf{w}) = \mu_\alpha h + (1-\alpha)^2\Psi_0(\mathbf{w}),$$

where $\mu_\alpha = (1 + \alpha^2)/2$.

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We now bound the ReLU remainder. Since u and q are orthonormal,

$$\|\Psi_0(\mathbf{w})\|^2 = \frac{b^2}{4\pi^2}\left[(\rho\theta - \sin\theta)^2 + \theta^2 s^2\right].$$

Also

$$\|\mathbf{w} - \mathbf{w}^\star\|^2 = (a - b\rho)^2 + b^2 s^2 \geq b^2 s^2.$$

The case $\theta = 0$ gives $\Psi_0(\mathbf{w}) = 0$ by continuity. For $\theta \in (0, \pi/2]$,

$$\frac{\|\Psi_0(\mathbf{w})\|^2}{\|\mathbf{w} - \mathbf{w}^\star\|^2} \leq \frac{1}{4\pi^2}\left[(\theta\cot\theta - 1)^2 + \theta^2\right].$$

It remains to use a scalar inequality. Let $g(\theta) = (1 - \theta\cot\theta)/\theta = 1/\theta - \cot\theta$. Then $g'(\theta) = \csc^2\theta - 1/\theta^2 \geq 0$ on $(0, \pi/2]$ because $\sin\theta \leq \theta$, and $g(0+) = 0$, $g(\pi/2) = 2/\pi$. Hence $0 \leq 1 - \theta\cot\theta \leq 2\theta/\pi$. Therefore

$$\|\Psi_0(\mathbf{w})\| \leq \frac{\theta}{2\pi}\sqrt{1 + \frac{4}{\pi^2}}\,\|\mathbf{w} - \mathbf{w}^\star\| = C_0\theta\,\|\mathbf{w} - \mathbf{w}^\star\|.$$

Multiplying by $(1-\alpha)^2$ and using $\theta \leq \theta_0$ gives the column bound. Squaring, summing over columns, and taking square roots proves

$$\|\Psi_\alpha(\mathbf{W})\|_{\mathrm{F}} \leq (1-\alpha)^2 C_0\theta_0 \left\|\mathbf{W} - \widetilde{\mathbf{W}}^\star\right\|_{\mathrm{F}}.$$

A.4 Proof of Lemma 4.4

The proof is for a fixed path independent of the sample. The fresh-batch stage statement follows by conditioning on the previous stages; after conditioning, the current iterate is fixed and the new batch is independent.

Fix t and j, write $h_{t,j} = w_{t,j} - w_j^\star$, and define the vector summand

$$Z_i^{t,j} = \left(\sigma_\alpha(x_i^\top w_{t,j}) - \sigma_\alpha(x_i^\top w_j^\star)\right)\sigma'_\alpha(x_i^\top w_{t,j})x_i.$$

Then

$$\nabla_{w_j}\mathcal{L}_N(\mathbf{W}_t) - \nabla_{w_j}\mathcal{L}_{\mathrm{pop}}(\mathbf{W}_t) = \frac{1}{N}\sum_{i=1}^N\left(Z_i^{t,j} - \mathbb{E}Z_i^{t,j}\right).$$

At the nondifferentiable point of σ_α, choose any derivative in $[\alpha, 1]$; all bounds below use only $|\sigma'_\alpha| \leq 1$.

Let $u \in \mathbb{S}^{d-1}$. The one-dimensional projection is bounded in absolute value by a product of Gaussian linear forms:

$$\begin{aligned}\left|u^\top Z_i^{t,j}\right| &\leq \left|u^\top x_i\right|\left|\sigma_\alpha(x_i^\top w_{t,j}) - \sigma_\alpha(x_i^\top w_j^\star)\right|\left|\sigma'_\alpha(x_i^\top w_{t,j})\right| \\ &\leq \left|u^\top x_i\right|\left|x_i^\top h_{t,j}\right|,\end{aligned}$$

because σ_α is 1-Lipschitz for $\alpha \in [0,1]$. Hence

$$\left\|\left|u^\top x\right|\left|x^\top h_{t,j}\right|\right\|_{\psi_1} \leq C\left\|u^\top x\right\|_{\psi_2}\left\|x^\top h_{t,j}\right\|_{\psi_2} \leq C\,\|h_{t,j}\|.$$

Centering increases the ψ_1 norm by at most a universal factor, so

$$\left\|u^\top\left(Z_i^{t,j} - \mathbb{E}Z_i^{t,j}\right)\right\|_{\psi_1} \leq C\,\|h_{t,j}\|.$$

For the variance, centering can only decrease the second moment:

$$\begin{aligned}\mathrm{Var}\left(u^\top Z_i^{t,j}\right) &\leq \mathbb{E}\left[(u^\top Z_i^{t,j})^2\right] \\ &\leq \mathbb{E}\left[(u^\top x_i)^2(x_i^\top h_{t,j})^2\right] \leq 3\,\|h_{t,j}\|^2,\end{aligned}$$

where the last inequality is Isserlis' theorem.

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Bernstein's inequality for centered sub-exponential variables now gives, for any $\tau > 0$, with probability at least $1 - 2e^{-\tau}$,

$$\left| \frac{1}{N} \sum_{i=1}^{N} u^\top \left(Z_i^{t,j} - \mathbb{E} Z_i^{t,j} \right) \right| \leq C \left[\|h_{t,j}\| \sqrt{\frac{\tau}{N}} + \|h_{t,j}\| \frac{\tau}{N} \right].$$

If $N \geq C_1 \tau$, the linear Bernstein term is absorbed into the square-root term:

$$\left| \frac{1}{N} \sum_{i=1}^{N} u^\top \left(Z_i^{t,j} - \mathbb{E} Z_i^{t,j} \right) \right| \leq C \|h_{t,j}\| \sqrt{\frac{\tau}{N}}.$$

Let $\mathcal{N}$ be a $1/2$-net of $\mathbb{S}^{d-1}$ with $|\mathcal{N}| \leq 5^d$, and set

$$\tau = \log \frac{2 \cdot 5^d m(T+1)}{\delta}.$$

Union bounding over $u \in \mathcal{N}$, all columns j, and all path points $t = 0, \ldots, T$ gives the preceding scalar bound simultaneously. For any $z \in \mathbb{R}^d$,

$$\|z\|_2 \leq 2 \sup_{u \in \mathcal{N}} \left| u^\top z \right|,$$

so, on the same event,

$$\left\| \nabla_{w_j} \mathcal{L}_N(\mathbf{W}_t) - \nabla_{w_j} \mathcal{L}_{\text{pop}}(\mathbf{W}_t) \right\|_2 \leq C \|h_{t,j}\| \sqrt{\frac{\log(2 \cdot 5^d m(T+1)/\delta)}{N}}.$$

Summing the column bounds in quadrature yields, for every t,

$$\|\nabla \mathcal{L}_N(\mathbf{W}_t) - \nabla \mathcal{L}_{\text{pop}}(\mathbf{W}_t)\|_{\text{F}} \leq C \left\| \mathbf{W}_t - \widetilde{\mathbf{W}}^\star \right\|_{\text{F}} \sqrt{\frac{\log(2 \cdot 5^d m(T+1)/\delta)}{N}}.$$

Finally,

$$\log \frac{2 \cdot 5^d m(T+1)}{\delta} \leq C d \log \left(\frac{10 d m(T+1)}{\delta} \right)$$

after enlarging the universal constant. Taking the maximum over t proves the stated bound.

A.5 Proof of Proposition 4.5 and statement and proof of Corollary A.1

We first prove Proposition 4.5, then state and prove Corollary A.1, which records the deterministic accounting one gets when a free-factor solver is compared to its direction-locked counterpart.

Proof of Proposition 4.5. Set

$$\widetilde{A}_k = -\nabla \mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k \rangle}) / \mu_\alpha.$$

Because $\widetilde{\mathbf{W}}^\star = \mathbf{W}_0 + \Delta$ and $\mathbf{W}^{\langle k \rangle} = \mathbf{W}_0 + \sum_{j<k} U_j$, we have

$$\mathbf{W}^{\langle k \rangle} - \widetilde{\mathbf{W}}^\star = -R_k^W.$$

Lemma 4.3 gives

$$-\nabla \mathcal{L}_{\text{pop}}(\mathbf{W}^{\langle k \rangle}) / \mu_\alpha = R_k^W - \Psi_\alpha(\mathbf{W}^{\langle k \rangle}) / \mu_\alpha.$$

On the concentration event of Lemma 4.4,

$$\begin{aligned} \left\| \widetilde{A}_k - R_k^W \right\|_{\text{F}} &\leq \frac{\left\| \Psi_\alpha(\mathbf{W}^{\langle k \rangle}) \right\|_{\text{F}}}{\mu_\alpha} + \frac{\left\| \nabla \mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k \rangle}) - \nabla \mathcal{L}_{\text{pop}}(\mathbf{W}^{\langle k \rangle}) \right\|_{\text{F}}}{\mu_\alpha} \\ &\leq \frac{(1-\alpha)^2 C_0 \theta_0 + \gamma_N}{\mu_\alpha} \left\| R_k^W \right\|_{\text{F}} = \eta_\alpha \left\| R_k^W \right\|_{\text{F}}. \end{aligned}$$

Therefore, with

$$e_k := \eta_\alpha \left\| R_k^W \right\|_{\text{F}} + \left\| R_k^W - \Delta_k^\star \right\|_{\text{F}},$$

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the hypothesis gives

$$\left\|\widetilde{A}_k - \Delta_k^\star\right\|_{\mathrm{F}} \le e_k \le \mathcal{T}_k^\star/4.$$

Write $E_k = \widetilde{A}_k - \Delta_k^\star$ and let $\sigma_{1,k} \ge \sigma_{2,k} \ge \cdots$ be the singular values of $\Delta_k^\star$. Let $\hat{\sigma}_{1,k} \ge \hat{\sigma}_{2,k} \ge \cdots$ be those of $\widetilde{A}_k$. Weyl's inequality gives, for every ℓ,

$$|\hat{\sigma}_{\ell,k} - \sigma_{\ell,k}| \le \|E_k\|_2 \le \|E_k\|_{\mathrm{F}}.$$

In particular,

$$\hat{\sigma}_{1,k} - \hat{\sigma}_{2,k} \ge (\sigma_{1,k} - \sigma_{2,k}) - 2\|E_k\|_2 \ge \mathcal{T}_k^\star/2,$$

and $\hat{\sigma}_{1,k} \ge \sigma_{1,k} - \|E_k\|_2 \ge 3\sigma_{1,k}/4$ whenever $\mathcal{T}_k^\star = \sigma_{1,k}$ is the active part of the minimum. Thus the leading singular direction remains simple and separated.

We now apply the rank-one form of Wedin's theorem. Equivalently, apply the Davis-Kahan theorem to the symmetric dilation

$$\mathcal{D}(M) = \begin{bmatrix} 0 & M \\ M^\top & 0 \end{bmatrix}.$$

The perturbation satisfies $\|\mathcal{D}(E_k)\|_2 = \|E_k\|_2 \le \|E_k\|_{\mathrm{F}}$, while the relevant eigenvalue gap of $\mathcal{D}(\Delta_k^\star)$ around its leading positive eigenvalue is at least $\mathcal{T}_k^\star = \min\{\sigma_{1,k} - \sigma_{2,k}, \sigma_{1,k}\}$. Since $\|E_k\|_2 \le \mathcal{T}_k^\star/4$, the one-dimensional left and right singular subspaces obey

$$\sin\angle(\hat{u}_k, u_{1,k}) + \sin\angle(\hat{v}_k, v_{1,k}) \le C\frac{\|E_k\|_{\mathrm{F}}}{\mathcal{T}_k^\star}.$$

Choosing the joint signs so that the products are compared consistently,

$$\|\hat{u}_k - u_{1,k}\|_2 \le 2\sin\angle(\hat{u}_k, u_{1,k}), \qquad \|\hat{v}_k - v_{1,k}\|_2 \le 2\sin\angle(\hat{v}_k, v_{1,k}).$$

Therefore

$$\begin{aligned}\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} &\le \left\|(\hat{u}_k - u_{1,k})\hat{v}_k^\top\right\|_{\mathrm{F}} + \left\|u_{1,k}(\hat{v}_k - v_{1,k})^\top\right\|_{\mathrm{F}} \\ &\le C\frac{\|E_k\|_{\mathrm{F}}}{\mathcal{T}_k^\star} \le C\frac{e_k}{\mathcal{T}_k^\star}.\end{aligned}$$

This proves the stated dyad bound.

It remains to convert dyad error into update error. The algorithm returns a scalar s_k satisfying

$$\varepsilon_{k,\mathrm{amp}} = \left|s_k - \sigma_1(\widetilde{A}_k)\right| = |s_k - \hat{\sigma}_{1,k}|.$$

Weyl's inequality also gives

$$|\hat{\sigma}_{1,k} - \sigma_{1,k}| \le \|E_k\|_2 \le e_k,$$

so

$$|s_k - \sigma_{1,k}| \le \varepsilon_{k,\mathrm{amp}} + e_k.$$

Using $U_k^\star = \sigma_{1,k}u_{1,k}v_{1,k}^\top$,

$$\begin{aligned}\|U_k - U_k^\star\|_{\mathrm{F}} &= \left\|s_k\hat{u}_k\hat{v}_k^\top - \sigma_{1,k}u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \\ &\le |s_k - \sigma_{1,k}| + \sigma_{1,k}\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \\ &\le \varepsilon_{k,\mathrm{amp}} + e_k + C\sigma_{1,k}\frac{e_k}{\mathcal{T}_k^\star}.\end{aligned}$$

Since $\mathcal{T}_k^\star \le \sigma_{1,k}$, the additive e_k term is absorbed into the last term after increasing the universal constant. This yields

$$\|U_k - U_k^\star\|_{\mathrm{F}} \le \varepsilon_{k,\mathrm{amp}} + C'\frac{e_k}{\mathcal{T}_k^\star}\sigma_1(\Delta_k^\star),$$

which is the claimed stage bound.

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Corollary A.1 (Accounting for a rotated rank-one stage solver)**.** *At stage k of an actual free-factor path, construct the reference direction-locked update $\bar{U}_k = s_k \hat{u}_k \hat{v}_k^\top$ from the same current iterate $\mathbf{W}^{\langle k \rangle}$ and the same stage batch $\mathcal{B}_k$ used by the free-factor solver. Suppose Proposition 4.5 applies at this iterate and batch. If the free-factor solver returns $\widetilde{U}_k = b_k a_k^\top$, define $\varepsilon_{k,\mathrm{rot}} = \left\| \widetilde{U}_k - \bar{U}_k \right\|_{\mathrm{F}}$. Then*

$$\left\| \widetilde{U}_k - U_k^\star \right\|_{\mathrm{F}} \le \varepsilon_{k,\mathrm{rot}} + \left\| \bar{U}_k - U_k^\star \right\|_{\mathrm{F}},$$

where the second term is bounded by Proposition 4.5. Consequently, if Proposition 4.5 holds at every stage along the actual free-factor path, Proposition 4.6 applies to the free-factor sequence with the additional sum $\sum_{k=1}^{r} \varepsilon_{k,\mathrm{rot}}$. This does not prove that $\varepsilon_{k,\mathrm{rot}}$ is small; it only makes the missing solver-stability term explicit. The hypothesis that Proposition 4.5 applies along the actual free-factor path is itself a precondition: free-factor iterates differ from direction-locked iterates, and we do not separately analyze when the sector and perturbation-budget conditions continue to hold along that path.

Proof of Corollary A.1. The result is deterministic. At a single stage, insert the direction-locked reference update $\bar{U}_k$ and use the triangle inequality:

$$\left\| \widetilde{U}_k - U_k^\star \right\|_{\mathrm{F}} \le \left\| \widetilde{U}_k - \bar{U}_k \right\|_{\mathrm{F}} + \left\| \bar{U}_k - U_k^\star \right\|_{\mathrm{F}}.$$

The first term is the definition of $\varepsilon_{k,\mathrm{rot}}$. The second term is exactly the direction-locked stage error controlled by Proposition 4.5. Repeating the same insertion for each stage and applying Proposition 4.6 gives the same additive end-to-end bound with the added term

$$\sum_{k=1}^{r} \varepsilon_{k,\mathrm{rot}}.$$

No probabilistic or optimization claim about the size of $\varepsilon_{k,\mathrm{rot}}$ is used.

A.6 Statement and proof of Proposition A.2

The same surrogate-to-dyad principle applies to smooth matched activations through oracle Hessian preconditioning. This is a local, analysis-level analogue: it requires oracle Hessian preconditioning, is not used in the all-stage theorem, and is included to scope the surrogate principle beyond Leaky-ReLU.

Proposition A.2 (Smooth matched Hessian-preconditioned surrogate)**.** *Consider the one-matrix teacher-student model*

$$y_j = \phi(x^\top \widetilde{w}_j^\star), \qquad \mathcal{L}(\mathbf{W}) = \frac{1}{2} \sum_{j=1}^{m} \mathbb{E}\left[\left(\phi(x^\top w_j) - \phi(x^\top \widetilde{w}_j^\star) \right)^2 \right],$$

with $\mathcal{L}_{\mathcal{B}}$ denoting the corresponding fresh-batch empirical loss. Assume the matched activation satisfies $|\phi'| \le B_1$ and ϕ' is B_2-Lipschitz. Let $R = \widetilde{\mathbf{W}}^\star - \mathbf{W}$, $r_j = \widetilde{w}_j^\star - w_j$, and assume

$$\sup_{\|a\|=\|b\|=\|u\|=1} \mathbb{E}\left| (a^\top x)(b^\top x)(u^\top x) \right| \le M_3.$$

Define

$$H_j^\star = \mathbb{E}[\phi'(x^\top \widetilde{w}_j^\star)^2 x x^\top], \qquad \mathcal{P}_\star(G) = [(H_1^\star)^{-1} g_1, \ldots, (H_m^\star)^{-1} g_m],$$

and suppose $H_j^\star \succeq \lambda_{\min} I$ for all j. If a fresh stage batch is drawn after $\mathbf{W}$ is fixed and $\left\| x^\top a \right\|_{\psi_2} \le K \|a\|$, then with probability at least $1 - \delta$,

$$\left\| \mathcal{P}_\star(-\nabla \mathcal{L}_{\mathcal{B}}(\mathbf{W})) - R \right\|_{\mathrm{F}} \le \left[\frac{2 B_1 B_2 M_3}{\lambda_{\min}} \max_j \|r_j\| + \frac{\gamma_{\mathrm{sm}}}{\lambda_{\min}} \right] \|R\|_{\mathrm{F}},$$

where

$$\gamma_{\mathrm{sm}} = C B_1^2 K^2 \left[\sqrt{\frac{dm + \log(2/\delta)}{|\mathcal{B}|}} + \frac{dm + \log(2/\delta)}{|\mathcal{B}|} \right].$$

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Proof of Proposition A.2. For a single column, write $w = w_j$, $\widetilde{w}^\star = \widetilde{w}_j^\star$, and $e = w - \widetilde{w}^\star = -r_j$. Since $|\phi'| \le B_1$, the loss is differentiable under the expectation by dominated convergence. With $z_t = x^\top(\widetilde{w}^\star + te)$, the fundamental theorem of calculus gives

$$\phi(x^\top w) - \phi(x^\top \widetilde{w}^\star) = \int_0^1 \phi'(z_t) x^\top e dt.$$

Hence

$$\nabla_{w_j} \mathcal{L}(\mathbf{W}) = H_j(\mathbf{W}, \widetilde{\mathbf{W}}^\star)(w_j - \widetilde{w}_j^\star),$$

where

$$H_j(\mathbf{W}, \widetilde{\mathbf{W}}^\star) = \mathbb{E}\left[xx^\top \int_0^1 \phi'(x^\top(\widetilde{w}_j^\star + t(w_j - \widetilde{w}_j^\star)))\phi'(x^\top w_j) dt\right].$$

For unit vectors a, b, Lipschitzness of ϕ' gives

$$\begin{aligned}
&\left|a^\top (H_j(\mathbf{W}, \widetilde{\mathbf{W}}^\star) - H_j^\star) b\right| \\
&\le \mathbb{E}\left[\left|a^\top x\right| \left|b^\top x\right| \int_0^1 \left|\phi'(z_t)\phi'(z_1) - \phi'(z_0)^2\right| dt\right] \\
&\le 2B_1 B_2 \mathbb{E}\left|(a^\top x)(b^\top x)(x^\top e)\right| \le 2B_1 B_2 M_3 \left\|e\right\|.
\end{aligned}$$

Thus

$$-\nabla_{w_j} \mathcal{L}(\mathbf{W}) = H_j^\star r_j - q_j, \qquad \left\|q_j\right\| \le 2B_1 B_2 M_3 \left\|r_j\right\|^2.$$

Applying $(H_j^\star)^{-1}$ columnwise and summing over j gives the population bound

$$\left\|\mathcal{P}_\star(-\nabla \mathcal{L}(\mathbf{W})) - R\right\|_{\mathrm{F}} \le \frac{2B_1 B_2 M_3}{\lambda_{\min}} \left(\max_j \left\|r_j\right\|\right) \left\|R\right\|_{\mathrm{F}}.$$

It remains to control the empirical gradient. For any fixed matrix $U = [u_1, \ldots, u_m]$ with $\left\|U\right\|_{\mathrm{F}} = 1$, the single-sample directional gradient satisfies

$$\left|\langle \nabla \ell_i(\mathbf{W}), U\rangle\right| \le B_1^2 \sum_{j=1}^m \left|x_i^\top r_j\right| \left|x_i^\top u_j\right|.$$

The product of two sub-Gaussian random variables is sub-exponential, so

$$\left\|(x_i^\top r_j)(x_i^\top u_j)\right\|_{\psi_1} \le CK^2 \left\|r_j\right\| \left\|u_j\right\|.$$

By the triangle inequality for ψ_1 norms and Cauchy–Schwarz over columns,

$$\left\|\langle \nabla \ell_i(\mathbf{W}) - \nabla \mathcal{L}(\mathbf{W}), U\rangle\right\|_{\psi_1} \le CB_1^2 K^2 \left\|R\right\|_{\mathrm{F}}.$$

Bernstein's inequality for each fixed U, followed by a $1/2$-net of the Frobenius unit sphere in $\mathbb{R}^{d\times m}$ with cardinality at most 5^{dm}, gives with probability at least $1 - \delta$,

$$\left\|\nabla \mathcal{L}_{\mathcal{B}}(\mathbf{W}) - \nabla \mathcal{L}(\mathbf{W})\right\|_{\mathrm{F}} \le \gamma_{\mathrm{sm}} \left\|R\right\|_{\mathrm{F}}.$$

Applying $\mathcal{P}_\star$ to this deviation costs a factor $\lambda_{\min}^{-1}$. Combining the empirical deviation with the population bound proves Proposition A.2. □

A.7 Proof of Proposition 4.6 and Proposition 4.7

Let

$$D_k = R_k^W - \Delta_k^\star.$$

Using

$$R_k^W = \Delta - \sum_{j<k} U_j, \qquad \Delta_k^\star = \Delta - \sum_{j<k} U_j^\star,$$

we get

$$D_k = -\sum_{j<k} (U_j - U_j^\star).$$

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After r stages,

$$\Delta - \widehat{\Delta}^{(r)} = \Delta^\star_{r+1} + D_{r+1}.$$

The triangle inequality gives

$$\left\|\Delta - \widehat{\Delta}^{(r)}\right\|_{\mathrm{F}} \le \left\|\Delta^\star_{r+1}\right\|_{\mathrm{F}} + \sum_{k=1}^{r} \|U_k - U_k^\star\|_{\mathrm{F}}\,.$$

It remains to identify the oracle tail. If $\Delta = \sum_{\ell\ge 1} \sigma_\ell u_\ell v_\ell^\top$ is an SVD, then successive exact best rank-1 deflation removes $\sigma_1 u_1 v_1^\top, \dots, \sigma_r u_r v_r^\top$ (with an arbitrary choice inside tied singular subspaces). Hence $\Delta^\star_{r+1} = \sum_{\ell>r} \sigma_\ell u_\ell v_\ell^\top$, and $\left\|\Delta^\star_{r+1}\right\|_{\mathrm{F}} = (\sum_{j>r} \sigma_j(\Delta)^2)^{1/2}$.

For any input x, the coordinatewise Leaky-ReLU map is 1-Lipschitz for $\alpha \in [0,1]$ in Euclidean norm:

$$\left\|\sigma_\alpha(x^\top W) - \sigma_\alpha(x^\top \widetilde{W}^\star)\right\|_2 \le \left\|x^\top (W - \widetilde{W}^\star)\right\|_2.$$

Taking expectation over $x \sim \mathcal{N}(0, I_d)$ gives

$$\mathcal{L}_{\mathrm{pop}}(W) \le \frac{1}{2}\mathbb{E}\left\|x^\top (W - \widetilde{W}^\star)\right\|_2^2 = \frac{1}{2}\left\|W - \widetilde{W}^\star\right\|_{\mathrm{F}}^2.$$

The last equality follows from

$$\mathbb{E}\left\|x^\top A\right\|_2^2 = \mathbb{E}\,\mathrm{tr}(A^\top x x^\top A) = \mathrm{tr}(A^\top A) = \|A\|_{\mathrm{F}}^2$$

with $A = W - \widetilde{W}^\star$.

A.8 Statement and proof of Corollary A.3

Corollary A.3 (Quantitative fixed-direction scalar refitting)**.** *Fix dyads $\widehat{D}_k = \hat{u}_k \hat{v}_k^\top$, $k = 1, \dots, r$, and define $\mathbf{W}(c) = \mathbf{W}_0 + \sum_{k=1}^r c_k \widehat{D}_k$. Let $s_k^\star = \sigma_1(\Delta_k^\star)$ and $D_k^\star = U_k^\star / s_k^\star$ when $s_k^\star > 0$; terms with $s_k^\star = 0$ are omitted. If $\hat{c}$ is an ε-optimal population scalar refit,*

$$\mathcal{L}_{\mathrm{pop}}(\mathbf{W}(\hat{c})) \le \inf_{c\in\mathbb{R}^r} \mathcal{L}_{\mathrm{pop}}(\mathbf{W}(c)) + \varepsilon,$$

then

$$\mathcal{L}_{\mathrm{pop}}(\mathbf{W}(\hat{c})) \le \frac{1}{2}\left(\left(\sum_{j>r}\sigma_j(\Delta)^2\right)^{1/2} + \sum_{k=1}^r s_k^\star \left\|\widehat{D}_k - D_k^\star\right\|_{\mathrm{F}}\right)^2 + \varepsilon.$$

An ε-optimal scalar refit therefore controls fitted loss for fixed directions; it does not imply better directions or smaller parameter error.

Proof of Corollary A.3. Let $s^\star = (s_1^\star, \dots, s_r^\star)$ be the oracle amplitude vector. By ε-optimality,

$$\mathcal{L}_{\mathrm{pop}}(\mathbf{W}(\hat{c})) \le \mathcal{L}_{\mathrm{pop}}(\mathbf{W}(s^\star)) + \varepsilon.$$

Using $\Delta = \Delta^\star_{r+1} + \sum_{k=1}^r s_k^\star D_k^\star$,

$$\begin{aligned}
\left\|\mathbf{W}(s^\star) - \widetilde{\mathbf{W}}^\star\right\|_{\mathrm{F}} &= \left\|\sum_{k=1}^r s_k^\star \widehat{D}_k - \Delta\right\|_{\mathrm{F}} \\
&= \left\|-\Delta^\star_{r+1} + \sum_{k=1}^r s_k^\star(\widehat{D}_k - D_k^\star)\right\|_{\mathrm{F}} \\
&\le \left\|\Delta^\star_{r+1}\right\|_{\mathrm{F}} + \sum_{k=1}^r s_k^\star \left\|\widehat{D}_k - D_k^\star\right\|_{\mathrm{F}}.
\end{aligned}$$

The first term is the oracle rank-r tail. Applying Proposition 4.7 to $\mathbf{W}(s^\star)$ gives the displayed population-loss bound. Finally, for any fixed scalar vector s, that vector is feasible for the same scalar-refit problem, so an ε-optimal refit for either $\mathcal{L}_{\mathrm{pop}}$ or an empirical refit-batch loss satisfies $\mathcal{L}(\mathbf{W}(\hat{c})) \le \mathcal{L}(\mathbf{W}(s)) + \varepsilon$ for that objective. The argument uses only feasibility of comparator scalars and does not assert that the fixed dyads themselves are accurate.

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A.9 Statement and proof of Theorem A.4

The conservative multi-stage certificate combines the per-stage perturbation bound (Proposition 4.5) with the additive propagation step (Proposition 4.6) into a closed end-to-end statement for the Gaussian Leaky-ReLU instantiation.

Theorem A.4 (Conservative all-stage error certificate)**.** *Assume $\mathcal{T}_k^\star > 0$ for $k \le r$, and assume the sector condition of Lemma 4.3 holds at every iterate $\mathbf{W}^{\langle k\rangle}$ with the same angle bound θ_0. For each stage, let $\gamma_k \ge 0$ and define*

$$E_k = \left\{ \left\| \nabla \mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k\rangle}) - \nabla \mathcal{L}_{\mathrm{pop}}(\mathbf{W}^{\langle k\rangle}) \right\|_{\mathrm{F}} \le \gamma_k \left\| R_k^W \right\|_{\mathrm{F}} \right\}.$$

Set

$$\eta_k = \frac{(1-\alpha)^2 C_0 \theta_0 + \gamma_k}{\mu_\alpha}, \quad s_k^\star = \sigma_1(\Delta_k^\star), \quad \rho_k^\star = \frac{s_k^\star}{\mathcal{T}_k^\star}, \quad \kappa_k^\star = \frac{\|\Delta_k^\star\|_{\mathrm{F}}}{\mathcal{T}_k^\star},$$

and let $\varepsilon_{k,\mathrm{amp}} = \left| s_k - \sigma_1(-\nabla \mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k\rangle})/\mu_\alpha) \right|$. Define $d_1 = 0$ and

$$d_{k+1} = \big(1 + C' \rho_k^\star (1+\eta_k)\big) d_k + \varepsilon_{k,\mathrm{amp}} + C' \eta_k \kappa_k^\star s_k^\star.$$

If, for every $k \le r$,

$$\eta_k \left(\kappa_k^\star + \frac{d_k}{\mathcal{T}_k^\star} \right) + \frac{d_k}{\mathcal{T}_k^\star} \le \frac{1}{4},$$

then on $E = \cap_{k=1}^r E_k$,

$$\left\| R_k^W - \Delta_k^\star \right\|_{\mathrm{F}} \le d_k, \qquad \|U_k - U_k^\star\|_{\mathrm{F}} \le d_{k+1} - d_k \quad (k \le r).$$

Consequently,

$$\left\| \Delta - \widehat{\Delta}^{(r)} \right\|_{\mathrm{F}} \le \left(\sum_{j>r} \sigma_j(\Delta)^2 \right)^{1/2} + d_{r+1},$$

and

$$\mathcal{L}_{\mathrm{pop}}(\mathbf{W}_0 + \widehat{\Delta}^{(r)}) \le \frac{1}{2} \left[\left(\sum_{j>r} \sigma_j(\Delta)^2 \right)^{1/2} + d_{r+1} \right]^2.$$

Reading the recurrence. The amplification factor $1 + C'\rho_k^\star(1+\eta_k)$ in front of d_k records how much past parameter error the current stage inherits, while the additive terms $\varepsilon_{k,\mathrm{amp}} + C'\eta_k\kappa_k^\star s_k^\star$ record what this stage contributes from scalar slack and nonlinearity bias. The sufficient condition $\eta_k(\kappa_k^\star + d_k/\mathcal{T}_k^\star) + d_k/\mathcal{T}_k^\star \le 1/4$ requires the combined nonlinearity-plus-sampling slack η_k and the relative past mismatch $d_k/\mathcal{T}_k^\star$ to remain a constant fraction of the current residual gap. Because past mismatch enters in Frobenius norm, a stage that consumes a sizeable fraction of its gap budget tightens every later stage's margin: this is why the certificate is conservative and not a generic convergence theorem.

Status of $\varepsilon_{k,\mathrm{amp}}$. The scalar slack $\varepsilon_{k,\mathrm{amp}} = |s_k - \sigma_1(A_k)|$ enters the recurrence as an additive per-stage term but is not bounded by Algorithm 1 in general. When $t_k = 0$, the algorithm sets $s_k = \sigma_1(A_k)$ and $\varepsilon_{k,\mathrm{amp}} = 0$ exactly, so the theorem applies with the slack removed. When $t_k > 0$, scalar refinement on $\mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k\rangle} + s_k \hat{u}_k \hat{v}_k^\top)$ may move s_k away from $\sigma_1(A_k)$; the theorem treats $\varepsilon_{k,\mathrm{amp}}$ as a per-stage measured quantity (reported by the diagnostic $|s_k - \sigma_1(A_k)|$ in Appendix B), and the budget β_k in Corollary A.6 accommodates any user-supplied tolerance. An analytical bound on $\varepsilon_{k,\mathrm{amp}}$ for the refined scalar is outside the scope of the local theorem and depends on the specific scalar solver.

Proof of Theorem A.4. For each stage let $D_k = R_k^W - \Delta_k^\star$. Since $R_1^W = \Delta = \Delta_1^\star$, we have $D_1 = 0 = d_1$. Work on $E = \cap_{k=1}^r E_k$ and suppose inductively that $\|D_k\|_{\mathrm{F}} \le d_k$. Then

$$\frac{\left\| R_k^W \right\|_{\mathrm{F}}}{\mathcal{T}_k^\star} \le \frac{\|\Delta_k^\star\|_{\mathrm{F}} + d_k}{\mathcal{T}_k^\star} = \kappa_k^\star + \frac{d_k}{\mathcal{T}_k^\star}, \qquad \frac{\left\| R_k^W - \Delta_k^\star \right\|_{\mathrm{F}}}{\mathcal{T}_k^\star} \le \frac{d_k}{\mathcal{T}_k^\star}.$$

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The envelope condition therefore implies the sufficient condition in Proposition 4.5. Applying that proposition at stage k gives

$$\begin{aligned}\|U_k - U_k^\star\|_{\mathrm{F}} &\le \varepsilon_{k,\mathrm{amp}} + C'\left[\eta_k\left(\kappa_k^\star + \frac{d_k}{\mathcal{T}_k^\star}\right) + \frac{d_k}{\mathcal{T}_k^\star}\right] s_k^\star \\ &= \varepsilon_{k,\mathrm{amp}} + C'\eta_k\kappa_k^\star s_k^\star + C'\rho_k^\star(1+\eta_k)d_k = d_{k+1} - d_k.\end{aligned}$$

Moreover,

$$D_{k+1} = R_{k+1}^W - \Delta_{k+1}^\star = (R_k^W - U_k) - (\Delta_k^\star - U_k^\star) = D_k - (U_k - U_k^\star),$$

so

$$\|D_{k+1}\|_{\mathrm{F}} \le \|D_k\|_{\mathrm{F}} + \|U_k - U_k^\star\|_{\mathrm{F}} \le d_k + (d_{k+1} - d_k) = d_{k+1}.$$

This closes the induction and proves the simultaneous residual-mismatch and stage-error bounds. The final parameter bound follows from Proposition 4.6 and the telescoping estimate

$$\sum_{k=1}^{r} \|U_k - U_k^\star\|_{\mathrm{F}} \le \sum_{k=1}^{r}(d_{k+1} - d_k) = d_{r+1}.$$

The population-prediction bound is then Proposition 4.7.

A.10 Statements and proofs of Corollaries A.5 and A.6

The first corollary specializes Theorem A.4 to a uniform fresh-batch setting and certifies sector control along the path; the second converts the per-stage feasibility condition into an observable oracle margin.

Corollary A.5 (Uniform fresh-batch all-stage bound with sector certification)**.** *Assume the setting of Theorem A.4 except possibly its sector condition. Suppose that, for all $k \le r$,*

$$\kappa_k^\star \le \kappa_\star, \quad \rho_k^\star \le \rho_\star, \quad s_k^\star \le s_\star, \quad \mathcal{T}_k^\star \ge \tau_\star, \quad \varepsilon_{k,\mathrm{amp}} \le \varepsilon_{\mathrm{amp}}.$$

Assume also

$$m_{\min} := \min_{j:\widetilde{w}_j^\star \neq 0} \left\|\widetilde{w}_j^\star\right\|_2 > 0.$$

Each stage uses an independent fresh batch of size N, with failure probability δ/r per stage. Let

$$\gamma = C_2\sqrt{\frac{d\log\left(\frac{10dmr}{\delta}\right)}{N}}, \qquad \eta = \frac{(1-\alpha)^2 C_0\theta_0 + \gamma}{\mu_\alpha},$$

and define $a = 1 + C'\rho_\star(1+\eta)$, $b = \varepsilon_{\mathrm{amp}} + C'\eta\kappa_\star s_\star$, and

$$\bar{d}_1 = 0, \qquad \bar{d}_{k+1} = a\bar{d}_k + b.$$

Equivalently,

$$\bar{d}_k = \begin{cases} b\dfrac{a^{k-1}-1}{a-1}, & a \neq 1, \\ (k-1)b, & a = 1. \end{cases}$$

If $N \ge C_1 d\log\left(\frac{10dmr}{\delta}\right)$ and

$$\eta\left(\kappa_\star + \frac{\bar{d}_r}{\tau_\star}\right) + \frac{\bar{d}_r}{\tau_\star} \le \frac{1}{4},$$

and

$$\max_{1\le k\le r}\left\{\|\Delta_k^\star\|_{\mathrm{F}} + \bar{d}_k\right\} \le m_{\min}\sin\theta_0,$$

then with probability at least $1-\delta$,

$$\left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}} \le \bar{d}_k \quad (k \le r), \qquad \left\|\Delta - \widehat{\Delta}^{(r)}\right\|_{\mathrm{F}} \le \left(\sum_{j>r}\sigma_j(\Delta)^2\right)^{1/2} + \bar{d}_{r+1},$$

and the same squared right-hand side times $1/2$ bounds $\mathcal{L}_{\mathrm{pop}}(\mathbf{W}_0 + \widehat{\Delta}^{(r)})$. The displayed maximum certifies the sector condition at every stage via Lemma C.1.

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Corollary A.6 (Stage-wise sample and sector certification budgets)**.** *Assume the setting of Theorem A.4. Let $\delta_1, \dots, \delta_r > 0$ with $\sum_k \delta_k \le \delta$, and fix scalar amplitude budgets $\beta_k \ge 0$. Define $d_1 = 0$ and, before stage k, set*

$$e_k = \frac{d_k}{\mathcal{T}_k^\star}, \qquad M_k = \frac{\mu_\alpha(1/4 - e_k)}{\kappa_k^\star + e_k} - (1-\alpha)^2 C_0 \theta_0.$$

If $M_k \le 0$, the all-stage certificate cannot certify another greedy direction at stage k from sample size alone. If $M_k > 0$ and the fresh stage batch size satisfies

$$N_k \ge C_1 d \log\left(\frac{10dm}{\delta_k}\right), \qquad N_k \ge C \frac{d \log\left(\frac{10dm}{\delta_k}\right)}{M_k^2},$$

then Lemma 4.4 gives the stage event E_k with $\gamma_k \le M_k$. If also $\varepsilon_{k,\mathrm{amp}} \le \beta_k$, update the envelope using

$$\eta_k = \frac{(1-\alpha)^2 C_0 \theta_0 + M_k}{\mu_\alpha}, \qquad d_{k+1} = \big(1 + C' \rho_k^\star (1 + \eta_k)\big) d_k + \beta_k + C' \eta_k \kappa_k^\star s_k^\star.$$

If the margin is positive at every stage, then with probability at least $1 - \delta$ the conclusions of Theorem A.4 hold with this envelope. Moreover, if $m_{\min} = \min_{j : \widetilde{w}_j^\star \neq 0} \left\|\widetilde{w}_j^\star\right\|_2 > 0$ and $\|\Delta_k^\star\|_{\mathrm{F}} + d_k \le m_{\min} \sin\theta_0$ for every stage, then the sector assumption in Theorem A.4 is certified by Lemma C.1. This is an oracle certification rule: it uses $\mathcal{T}_k^\star, \kappa_k^\star, \rho_k^\star$, and d_k, and is not an observable optimal stopping theorem.

Proofs. At each stage, apply Lemma 4.4 conditionally at the fixed current iterate with a fresh batch and failure probability δ/r. Since $N \ge C_1 d \log\big(\frac{10dmr}{\delta}\big)$, the concentration events

$$E_k = \left\{ \left\| \nabla \mathcal{L}_{\mathcal{B}_k}(\mathbf{W}^{\langle k \rangle}) - \nabla \mathcal{L}_{\mathrm{pop}}(\mathbf{W}^{\langle k \rangle}) \right\|_{\mathrm{F}} \le \gamma \left\| R_k^W \right\|_{\mathrm{F}} \right\}$$

hold simultaneously with probability at least $1 - \delta$, with

$$\gamma = C_2 \sqrt{\frac{d \log\big(\frac{10dmr}{\delta}\big)}{N}}.$$

Work on this event. We prove by induction that

$$\left\| R_k^W - \Delta_k^\star \right\|_{\mathrm{F}} \le \bar{d}_k \qquad (k = 1, \dots, r).$$

The base case holds because $R_1^W = \Delta = \Delta_1^\star$ and $\bar{d}_1 = 0$. Assume the claim holds at stage k. Then

$$\left\| R_k^W \right\|_{\mathrm{F}} \le \|\Delta_k^\star\|_{\mathrm{F}} + \bar{d}_k \le m_{\min} \sin\theta_0,$$

where the last inequality is the displayed sector-certification assumption. By Lemma C.1, the sector condition of Lemma 4.3 holds at $\mathbf{W}^{\langle k \rangle}$. Also

$$\frac{\left\| R_k^W \right\|_{\mathrm{F}}}{\mathcal{T}_k^\star} \le \kappa_k^\star + \frac{\bar{d}_k}{\mathcal{T}_k^\star} \le \kappa_\star + \frac{\bar{d}_r}{\tau_\star}, \qquad \frac{\left\| R_k^W - \Delta_k^\star \right\|_{\mathrm{F}}}{\mathcal{T}_k^\star} \le \frac{\bar{d}_r}{\tau_\star}.$$

Thus the displayed uniform condition implies the sufficient condition in Proposition 4.5. Applying Proposition 4.5 with $\eta_k = \eta$ gives

$$\|U_k - U_k^\star\|_{\mathrm{F}} \le \varepsilon_{\mathrm{amp}} + C' \eta \kappa_\star s_\star + C' \rho_\star (1 + \eta) \bar{d}_k = \bar{d}_{k+1} - \bar{d}_k.$$

Since

$$R_{k+1}^W - \Delta_{k+1}^\star = (R_k^W - \Delta_k^\star) - (U_k - U_k^\star),$$

we get

$$\left\| R_{k+1}^W - \Delta_{k+1}^\star \right\|_{\mathrm{F}} \le \bar{d}_k + (\bar{d}_{k+1} - \bar{d}_k) = \bar{d}_{k+1}.$$

This closes the simultaneous induction: before each application of Proposition 4.5, sector control has already been certified. The parameter-error bound follows from Proposition 4.6 and the telescoping sum

$$\sum_{k=1}^{r} \|U_k - U_k^\star\|_{\mathrm{F}} \le \sum_{k=1}^{r} (\bar{d}_{k+1} - \bar{d}_k) = \bar{d}_{r+1}.$$

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Finally, Proposition 4.7 gives the population-loss bound.

At stage k, Lemma 4.4 applied conditionally at the fixed current iterate with failure probability δ_k gives

$$\gamma_{N_k} \le C_2 \sqrt{\frac{d \log\left(\frac{10dm}{\delta_k}\right)}{N_k}},$$

provided $N_k \ge C_1 d \log\left(\frac{10dm}{\delta_k}\right)$. The displayed lower bound on N_k and a large enough universal constant C ensure $\gamma_{N_k} \le M_k$. Therefore

$$\eta_k \le \frac{(1-\alpha)^2 C_0 \theta_0 + M_k}{\mu_\alpha}.$$

By the definition of M_k and $e_k = d_k / \mathcal{T}_k^\star$,

$$\frac{(1-\alpha)^2 C_0 \theta_0 + M_k}{\mu_\alpha} (\kappa_k^\star + e_k) + e_k = \frac{1}{4}.$$

Hence the envelope condition in Theorem A.4 holds at stage k. If $\varepsilon_{k,\mathrm{amp}} \le \beta_k$, the displayed recurrence is an upper-envelope version of the recurrence in Theorem A.4. Repeating this argument stage by stage and union bounding $\sum_k \delta_k \le \delta$ gives the event $\cap_k E_k$ with probability at least $1-\delta$. Theorem A.4 then gives the claimed all-stage residual, parameter-error, and population-loss conclusions. If $M_k \le 0$, then even setting the sampling term γ_k to zero leaves no positive margin in the sufficient condition, so this proof cannot certify the next stage from more samples alone. The final sector-certification statement is exactly Lemma C.1 applied with $\left\|R_k^W\right\|_{\mathrm{F}} \le \|\Delta_k^\star\|_{\mathrm{F}} + d_k$ on the certified envelope.

B Synthetic Theory Diagnostics

B.1 Reproducibility details and synthetic setup

We run two synthetic suites. The quick suite uses $d = 40$, $m = 32$, $r = 5$, $N = 4096$, three seeds, and a total budget of 180 scalar-solver steps for the direction-locked method. The large suite uses $d = 128$, $m = 96$, $r = 8$, $N = 32768$, eight seeds, and a total budget of 1200 scalar-solver steps with at least 25 steps per stage. We use $\alpha = 0.05$ throughout, except in the nonlinearity sweep where we vary α explicitly. The sequential runs use fresh stage batches. We report training loss on a fixed reporting batch and prediction loss on a held-out test batch.

For scalar refinement we run full-batch Adam on a single amplitude per stage with learning rate 0.025, PyTorch default betas and epsilon, and a fresh optimizer state at every stage. Simultaneous baselines initialise random factors i.i.d. at scale 10^{-2} and train without weight decay. The scalar-solver ablation (Figure 9) uses the large mild-spectrum setting with the same seeds and stage batches as the main run.

We generate teacher spectra from three singular-value profiles: steep, $s_i \propto e^{-0.65(i-1)}$; mild, $s_i \propto e^{-0.22(i-1)}$; flat, $s_i \propto 1 - 0.025(i-1)$. We normalise so that $\|s\|_2 = 2$ in the large suite. In the rank-mismatch sweep we use learner ranks $\{2, 4, 6, 8, 10, 12\}$ against a rank-8 teacher. Ranks above 8 are over-specified and not covered by the local oracle-gap condition, since the teacher residual has been exhausted. In the nonlinearity sweep (Figure 1b) we use $N = 65536$ and vary $\alpha \in \{0, 0.3, 0.6, 0.9\}$ and $\|\Delta\|_{\mathrm{F}} \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$, reusing the eight seeds of the large suite.

B.2 Certified regime and local mechanism diagnostics

Throughout this appendix we mark the sufficient $1/4$ perturbation threshold of Proposition 4.5 as a dashed line. We label teacher-spectrum profiles steep, mild, and flat, as in Section B.1, and flat is the hardest case because the top dyad is least stable when residual singular-value gaps are small.

Before stress-testing the rank-8 run, we verify in Figure 3 that the local sufficient condition is observable in a deliberately favourable regime: a separated rank-3 teacher with N=131072. The empirical perturbation ratio stays below the $1/4$ threshold at every stage, and the sector and scalar diagnostics remain small.

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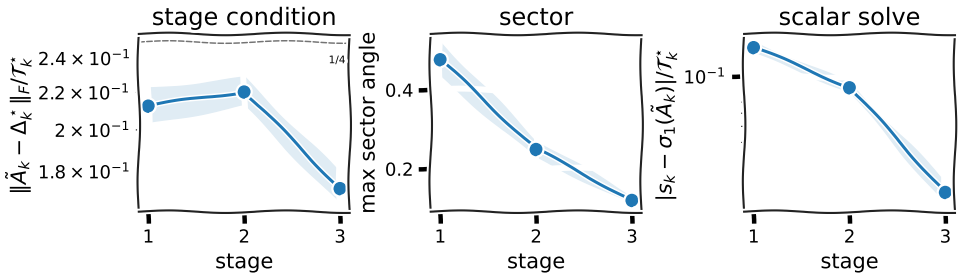


Figure 3: Calibration in the certified regime: separated rank-3 teacher with $N{=}131072$. Per-stage perturbation ratio (left), sector angle (middle), and scalar mismatch (right) for steep/mild/flat spectra. The empirical ratio stays below the $1/4$ sufficient threshold of Proposition 4.5 at every stage, so the local condition is satisfied throughout this run; this is the cleanly certified diagnostic evidence and is the regime in which the theorem applies. The harder rank-8 run in Figure 2 is outside the conservative sufficient regime in much of training and is reported as diagnostic evidence rather than as certification.

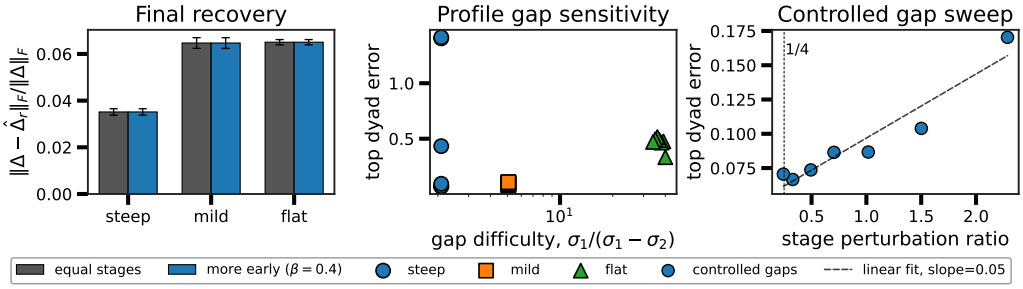


Figure 4: Spectral-gap sensitivity. Steep/mild/flat denote increasingly less separated teacher singular spectra. Left: final relative parameter error $\|\Delta - \widehat{\Delta}_r\|_{\mathrm{F}}/\|\Delta\|_{\mathrm{F}}$ on the rank-8 run is larger for flatter spectra, consistent with smaller residual gaps making each stage's top dyad less stable. Middle: top dyad error grows with the inverse residual relative gap $\sigma_1/(\sigma_1 - \sigma_2)$ across the three spectra. Right: in a controlled sweep that varies only the residual spectral gap (problem size, activation, sample size, and residual norm fixed), the top dyad error grows roughly linearly with the stage perturbation ratio $\|\widetilde{A} - \Delta^\star\|_{\mathrm{F}}/\mathcal{T}^\star$; the dashed vertical line marks the $1/4$ sufficient threshold of Proposition 4.5. The figure is consistent with the gap-normalized perturbation prediction of Theorem 4.1/Proposition 4.5 and with the sharpness statement of Proposition 4.2; it is not a global convergence statement.

We then return to the rank-8 run and track every per-stage quantity that enters Theorem A.4, not only the sufficient ratio of Figure 2. Figure 5 reports the accumulated past-stage mismatch d_k, the residual-to-gap factor $\kappa_k^\star$, the sector angle, the observed dyad error, and the scalar slack. The panels make each per-stage quantity in the recurrence visible; the all-stage envelope of Theorem A.4 certifies a stage only at those stages where the displayed feasibility condition holds.

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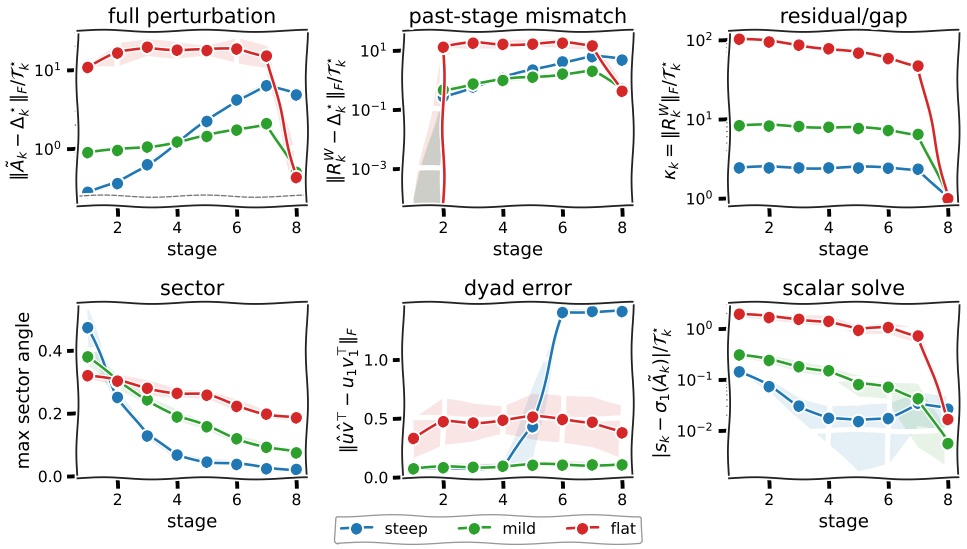


Figure 5: Expanded per-stage diagnostics on the rank-8 run. From left to right: empirical perturbation ratio (the sufficient quantity in Proposition 4.5); accumulated past-stage mismatch d_k from the recurrence in Theorem A.4; residual-to-gap factor $\kappa_k^\star = \|\Delta_k^\star\|_{\mathrm{F}}/\mathcal{T}_k^\star$, which controls the recurrence amplification; maximum sector angle, the assumption used by Lemma 4.3; observed top-dyad error; and scalar slack $\varepsilon_{k,\mathrm{amp}}$. The panels make each per-stage quantity in the theorem chain visible under the algorithm.

B.3 Sample dependence and end-to-end accounting

Sample size and learner rank enter the theorem at different points. The fresh batch size N enters through the concentration rate γ_N of Lemma 4.4, and the learner rank r enters through the oracle rank-r tail in Proposition 4.6. We test the two effects separately.

In Figure 6 we sweep N on the rank-8 run. Larger N tightens the maximum-over-stages perturbation ratio and improves both final parameter recovery and held-out prediction loss. The observed log-log slope is in qualitative agreement with the $N^{-1/2}$ guide of Lemma 4.4, since final error also contains nonlinear bias, gap effects, scalar slack, and accumulated past-stage mismatch.

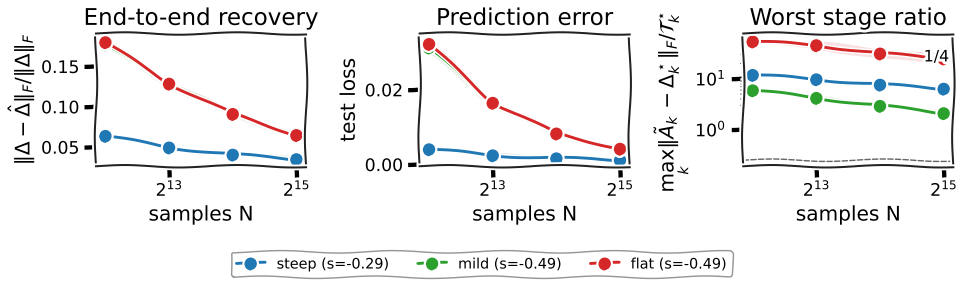


Figure 6: End-to-end sample scaling on the rank-8 run for steep/mild/flat teacher spectra. Increasing the fresh stage-batch size N improves final parameter recovery (left) and held-out prediction loss (middle); the right panel reports the maximum-over-stages perturbation ratio $\max_k \|\widetilde{A}_k - \Delta_k^\star\|_{\mathrm{F}}/\mathcal{T}_k^\star$, with the dashed $1/4$ line of Proposition 4.5. Log-log slope annotations compare the observed scaling against the $N^{-1/2}$ sampling guide of Lemma 4.4; agreement is qualitative, since final error also includes nonlinear bias, gap effects, scalar slack, and accumulated past-stage mismatch (Theorem A.4). The panels are jointly consistent with the theory chain: larger N tightens gradient concentration (Lemma 4.4), which improves stage-wise dyad recovery (Proposition 4.5) and propagates to final parameter and prediction error through Proposition 4.6/Proposition 4.7.

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In Figure 7 we sweep the learner rank r on the same teacher. We plot the learner error, the oracle rank-r tail, and the Proposition 4.6 accounting quantity (oracle tail plus the sum of measured per-stage errors) on a single axis. On the rank-≤ 8 side the learner sits at or below the accounting curve, as the additive bound predicts. Ranks $r > 8$ are over-specified: the oracle residual is exhausted, $\mathcal{T}_k^\star = 0$, and the local gap condition no longer certifies new dyads.

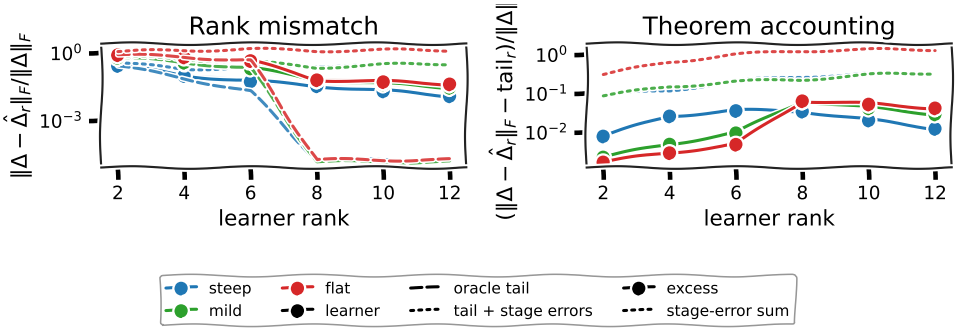


Figure 7: Rank mismatch against a rank-8 teacher (steep/mild/flat spectra). Left: final relative parameter error $\|\Delta - \widehat{\Delta}_r\|_{\mathrm{F}}/\|\Delta\|_{\mathrm{F}}$ versus learner rank r. Dashed: normalized oracle rank-r truncated-SVD tail $\mathrm{tail}_r/\|\Delta\|_{\mathrm{F}}$ with $\mathrm{tail}_r = (\sum_{j>r}\sigma_j(\Delta)^2)^{1/2}$. Solid: the direction-locked learner. Dotted: the Proposition 4.6 accounting quantity, oracle tail plus the sum of measured per-stage errors. Right: extra error above the oracle tail, $(\|\Delta - \widehat{\Delta}_r\|_{\mathrm{F}} - \mathrm{tail}_r)/\|\Delta\|_{\mathrm{F}}$, alongside the accumulated measured stage-error term. Under Proposition 4.6 the learner is bounded by the dotted curve, which is what the figure shows on the rank-≤ 8 side. Ranks > 8 are over-specified diagnostics: all nonzero oracle components have already been removed, $\mathcal{T}_k^\star = 0$, and the local gap condition no longer certifies new dyads.

B.4 Nonlinearity, ablations, and failure modes

The remaining synthetic figures probe how the local mechanism behaves when we vary parts of the setup that the theory holds fixed.

In Figure 8 we sweep the Leaky-ReLU slope α in the full sequential algorithm, not only in the stage-1 gradient check. Parameter recovery is fairly stable across slopes in this controlled setting, and prediction error increases as the teacher becomes nearly linear with fixed $\|\Delta\|_{\mathrm{F}}$. The mean perturbation ratio decreases with α, in line with the $(1-\alpha)^2$ scaling of the nonlinear remainder term in Lemma 4.3. In the rank-8 mild-spectrum configuration the ratio stays above the $1/4$ threshold, so the trend supports the predicted dependence without certifying the sufficient condition globally.

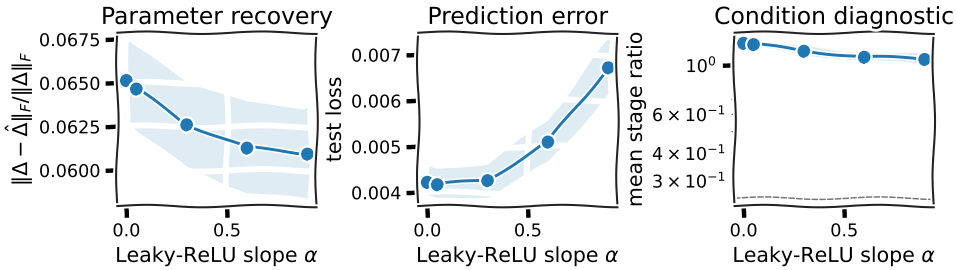


Figure 8: End-to-end recovery as the activation becomes more linear (rank-8 mild spectrum). Left/middle: final parameter and prediction error across $\alpha \in [0, 1)$. Right: empirical perturbation ratio averaged over stages; the dashed $1/4$ line is the sufficient threshold of Proposition 4.5. The mean ratio decreases with α, consistent with the $(1-\alpha)^2$ scaling of the nonlinear remainder term in Lemma 4.3, but stays above $1/4$ in this configuration. The trend is consistent with the predicted dependence; it does not certify the sufficient condition globally.

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In Figure 9 we ablate the one-dimensional scalar solver under the direction lock. In the large mild-spectrum run, pure spectral amplitude gives relative parameter error 0.0806 ± 0.0029. Exact piecewise-quadratic line search, scalar Adam, and backfitting all previous scalars are nearly indistinguishable: 0.0654 ± 0.0015, 0.0654 ± 0.0015, and 0.0653 ± 0.0015. Orthogonalising new directions does not help (0.0667 ± 0.0017). The per-stage slack $\varepsilon_{k,\mathrm{amp}}$ in Proposition 4.5 is therefore dominated by the recovered direction rather than by the choice of solver. The free-factor stage heuristic is more accurate (0.0106 ± 0.0004) but rotates away from the locked dyads with $\varepsilon_{\mathrm{rot}} \approx 0.068$ on average, and it is covered by the theorem only through Corollary A.1.

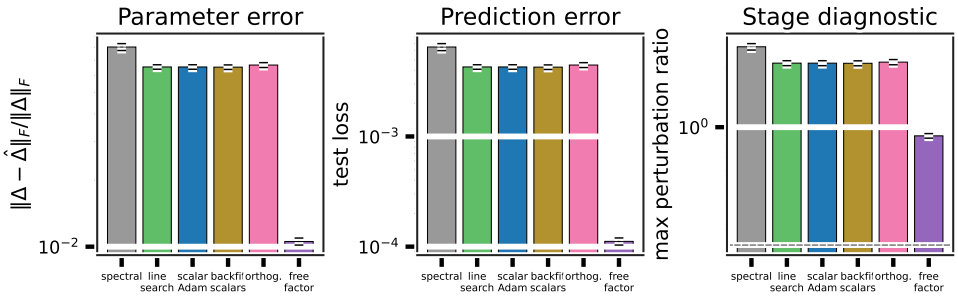


Figure 9: Scalar-solver ablation under the direction lock (large mild-spectrum run). Spectral amplitude, exact piecewise-quadratic line search, scalar Adam, and backfitting all previous scalars give nearly identical final errors, indicating that the per-stage slack $\varepsilon_{k,\mathrm{amp}}$ in Proposition 4.5 is dominated by the recovered direction rather than by the choice of one-dimensional solver. The free-factor heuristic is shown for context: it is more accurate but rotates away from the locked dyads, and Corollary A.1 only records the resulting rotation term $\varepsilon_{k,\mathrm{rot}}$ as a deterministic diagnostic without bounding it.

In Figure 10 we keep the same direction-locked algorithm and vary five axes that the theorem holds fixed: dimension, target rank, output noise, activation mismatch, and correlated inputs. We use the figure as a failure-mode diagnostic, not as additional evidence for a stronger theorem. Recovery degrades when the perturbation ratio grows, when the residual gaps shrink, or when the data geometry moves away from the isotropic Gaussian setting, in agreement with the local analysis.

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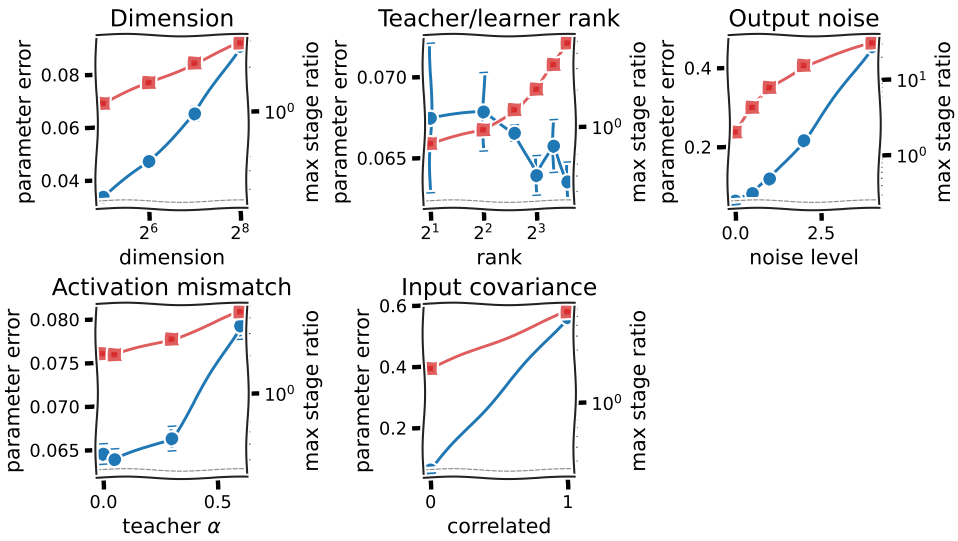


Figure 10: Synthetic robustness stress tests, outside the clean theorem assumptions. Final parameter error and the maximum empirical stage-condition ratio as we vary dimension, target rank, output noise, activation mismatch, and input covariance one axis at a time. The figure is a failure-mode diagnostic for the assumptions of Lemma 4.3/Proposition 4.5: recovery degrades when the perturbation ratio grows, residual gaps shrink, or the data geometry leaves the isotropic Gaussian setting. It is not additional evidence for a stronger theorem.

Finally, in Figure 11 we compare the analysed direction-locked variants (t_k=0 spectral and scalar-refined) against three baselines. The free-factor sequential method shares the SVD initialisation but then optimises both factors without a direction lock, and is covered by the theorem only through the rotation term of Corollary A.1. The last two baselines train a simultaneous rank-r factorisation $\Delta = BA^\top$ for the same Adam budget, from spectral and from random initialisation, and lie outside the per-stage theorem. We include this comparison for empirical context, not as additional theorem evidence.

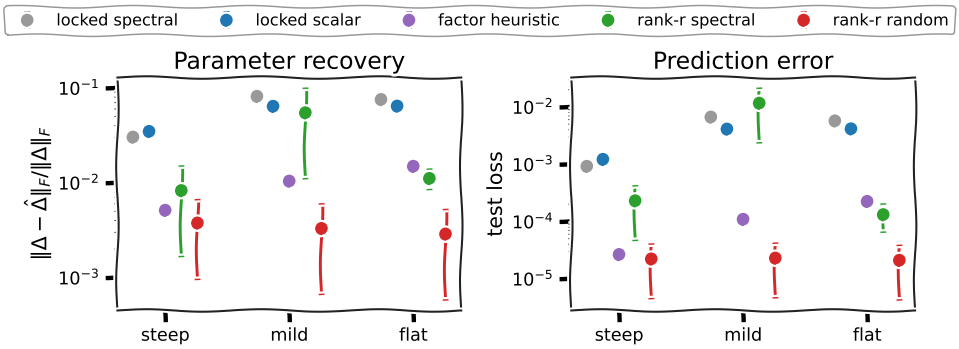


Figure 11: Baseline comparison on the large run. The two direction-locked variants (t_k=0 spectral and scalar-refined) are the methods analyzed by Proposition 4.5. The free-factor sequential baseline shares the same SVD initialization but optimizes both factors per stage; it is covered by the theorem only through the additive rotation term $\varepsilon_{k,\mathrm{rot}}$ of Corollary A.1. Simultaneous rank-r training, from spectral or random initialization, is shown for empirical context and is outside the stage perturbation theorem.

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C Auxiliary Theory and Diagnostics

C.1 Deterministic sector check

The sector assumption in Lemma 4.3 is not enforced by Algorithm 1. The following elementary check records when it is implied by ordinary parameter closeness.

Lemma C.1 (Parameter error implies sector control)**.** *Let $\widetilde{w}_j^\star$ denote the jth column of $\widetilde{W}^\star$, and assume $m_{\min} = \min_{j:\widetilde{w}_j^\star \neq 0} \|\widetilde{w}_j^\star\|_2 > 0$. If, for some $\theta_0 \in (0, \pi/2)$,*

$$\left\|W - \widetilde{W}^\star\right\|_{\mathrm{F}} \leq m_{\min} \sin\theta_0,$$

then every learner column corresponding to a nonzero teacher column is nonzero and satisfies $\angle(w_j, \widetilde{w}_j^\star) \leq \theta_0$. Consequently, along a stage-wise envelope it is sufficient that

$$\|\Delta_k^\star\|_{\mathrm{F}} + d_k \leq m_{\min} \sin\theta_0$$

to certify the sector condition at stage k, since $\left\|W^{\langle k\rangle} - \widetilde{W}^\star\right\|_{\mathrm{F}} = \left\|R_k^W\right\|_{\mathrm{F}} \leq \|\Delta_k^\star\|_{\mathrm{F}} + d_k$.

Proof. Fix a column with $\widetilde{w}_j^\star \neq 0$. The Frobenius bound implies $\|w_j - \widetilde{w}_j^\star\|_2 \leq m_{\min} \sin\theta_0 \leq \|\widetilde{w}_j^\star\|_2 \sin\theta_0 < \|\widetilde{w}_j^\star\|_2$, so $w_j \neq 0$. The distance from $\widetilde{w}_j^\star$ to the line spanned by w_j is $\|\widetilde{w}_j^\star\|_2 \sin\angle(w_j, \widetilde{w}_j^\star)$, and this distance is at most $\|w_j - \widetilde{w}_j^\star\|_2$. Also $\|w_j - \widetilde{w}_j^\star\|_2 < \|\widetilde{w}_j^\star\|_2$ implies $w_j^\top \widetilde{w}_j^\star > 0$, so the angle lies in $[0, \pi/2)$. Therefore

$$\sin\angle(w_j, \widetilde{w}_j^\star) \leq \frac{\|w_j - \widetilde{w}_j^\star\|_2}{\|\widetilde{w}_j^\star\|_2} \leq \sin\theta_0.$$

Since both angles lie in $[0, \pi/2]$, the sector bound follows. The final claim uses $R_k^W = \widetilde{W}^\star - W^{\langle k\rangle}$ and $R_k^W = \Delta_k^\star + (R_k^W - \Delta_k^\star)$. □

C.2 Independent-batch dyad stability

The following diagnostic is not used as a main contribution, but it is a useful check on whether independent fresh batches agree on the next dyad.

Corollary C.2 (Independent-batch top-dyad stability)**.** *Fix a stage k after $\mathbf{W}^{\langle k\rangle}$ has been fixed, and let $\mathcal{B}_k^{(1)}, \ldots, \mathcal{B}_k^{(M)}$ be independent fresh micro-batches. Define*

$$A_k^{(m)} := -\nabla \mathcal{L}_{\mathcal{B}_k^{(m)}}(\mathbf{W}^{\langle k\rangle})/\mu_\alpha, \qquad D_k^{(m)} := \hat{u}_k^{(m)} \hat{v}_k^{(m)\top}.$$

Let $D_k^\star = u_{1,k} v_{1,k}^\top$ be the top singular dyad of $\Delta_k^\star$. Suppose that Proposition 4.5 applies to each micro-batch and gives $\left\|D_k^{(m)} - D_k^\star\right\|_{\mathrm{F}} \leq \delta_k^{(m)}$. Then, for all $m \neq \ell$,

$$\left\|D_k^{(m)} - D_k^{(\ell)}\right\|_{\mathrm{F}} \leq \delta_k^{(m)} + \delta_k^{(\ell)}.$$

Consequently,

$$I_k(M) := \frac{2}{M(M-1)} \sum_{1 \leq m < \ell \leq M} \left\|D_k^{(m)} - D_k^{(\ell)}\right\|_{\mathrm{F}} \leq \frac{2}{M} \sum_{m=1}^{M} \delta_k^{(m)}.$$

In particular, if $\delta_k^{(m)} \leq \bar{\delta}_k$ for all m, then $I_k(M) \leq 2\bar{\delta}_k$.

Proof. Insert the oracle dyad and use the triangle inequality:

$$\left\|D_k^{(m)} - D_k^{(\ell)}\right\|_{\mathrm{F}} \leq \left\|D_k^{(m)} - D_k^\star\right\|_{\mathrm{F}} + \left\|D_k^{(\ell)} - D_k^\star\right\|_{\mathrm{F}} \leq \delta_k^{(m)} + \delta_k^{(\ell)}.$$

Averaging over unordered pairs gives the displayed bound, because each $\delta_k^{(m)}$ appears in exactly $M - 1$ pairs. □

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C.3 Diagnostic explanation of the stage condition

For equal-size micro-batches of size n, Proposition 4.5 gives, on the corresponding fresh-batch concentration events,

$$\delta_k^{(m)} \le C \frac{\eta_\alpha(\theta_0, \gamma_n) \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}}{\mathcal{T}_k^\star}.$$

Thus $I_k(M)$ inherits the same sampling, nonlinearity, past-error, and gap dependence as the one-stage perturbation term. The converse is not claimed: small $I_k(M)$ does not prove that Proposition 4.5 holds, since several batches can agree on the same wrong direction.

D Frozen-Feature Squared-Loss Bridge

D.1 Setup and motivation for the bridge

This appendix is a secondary frozen-feature squared-loss bridge. It isolates a one-matrix, frozen-representation setting and is not a theorem for deep-network LoRA, cross-entropy, or representation drift.

D.2 Deterministic frozen-feature bridge proposition

Proposition D.1 (Frozen-feature squared-loss bridge)**.** *Let $z \in \mathbb{R}^d$ be a frozen feature vector with covariance $\Sigma = \mathbb{E}[zz^\top]$, and consider the noiseless linear-head teacher $y = z^\top(\mathbf{W}_0 + \Delta)$. For the squared loss*

$$\mathcal{L}_{\mathrm{ff}}(\mathbf{W}) = \frac{1}{2}\mathbb{E}\left\|z^\top \mathbf{W} - z^\top(\mathbf{W}_0 + \Delta)\right\|_2^2,$$

define $R_k^W = \Delta - \sum_{j<k} U_j$ as before. Then

$$-\nabla \mathcal{L}_{\mathrm{ff}}(\mathbf{W}^{\langle k\rangle}) = \Sigma R_k^W.$$

For a fresh empirical batch $\mathcal{B}$, the same identity holds with Σ replaced by $\widehat{\Sigma}_{\mathcal{B}} = |\mathcal{B}|^{-1}\sum_{z_i \in \mathcal{B}} z_i z_i^\top$. Consequently,

$$\left\|-\nabla \mathcal{L}_{\mathrm{ff}}(\mathbf{W}^{\langle k\rangle}) - R_k^W\right\|_{\mathrm{F}} \le \|\Sigma - I\|_2 \left\|R_k^W\right\|_{\mathrm{F}},$$

and similarly with $\widehat{\Sigma}_{\mathcal{B}}$ for the empirical gradient.

In particular, if the features are whitened so that $\Sigma = I$, then the population negative gradient equals the current residual exactly. The population-gradient squared-loss analogue of Algorithm 1, with the same direction-locked extraction rule and exact scalar amplitudes $s_k = \sigma_1(R_k^W)$, recovers the rank-r truncated SVD of Δ:

$$\sum_{k=1}^{r} U_k = \Delta_{(r)}, \qquad \left\|\Delta - \sum_{k=1}^{r} U_k\right\|_{\mathrm{F}} = \left(\sum_{j>r} \sigma_j(\Delta)^2\right)^{1/2}.$$

The same conclusion holds if the selected SVD dyads are kept fixed and their scalar amplitudes are jointly refit by least squares. More generally, for any fixed set of dyads, joint scalar refitting cannot increase the squared-loss objective relative to the stagewise scalar values on those same dyads.

D.3 Finite-sample frozen-feature bridge proposition

Proposition D.2 (Finite-sample frozen-feature bridge)**.** *In the setting of Proposition D.1, assume $\mathcal{T}_k^\star > 0$ and assume the frozen features are population-whitened and sub-Gaussian: $\mathbb{E}[zz^\top] = I$ and $\|z\|_{\psi_2} \le K$. At a fixed stage k, draw a fresh batch $\mathcal{B}_k$ of size N after $\mathbf{W}^{\langle k\rangle}$ is fixed, and set*

$$\widetilde{A}_k = -\nabla \mathcal{L}_{\mathrm{ff},\mathcal{B}_k}(\mathbf{W}^{\langle k\rangle}) = \widehat{\Sigma}_{\mathcal{B}_k} R_k^W.$$

There is a constant C_K depending only on K such that, with probability at least $1-\delta$,

$$\left\|\widetilde{A}_k - R_k^W\right\|_{\mathrm{F}} \le \gamma_{\mathrm{ff}} \left\|R_k^W\right\|_{\mathrm{F}}, \qquad \gamma_{\mathrm{ff}} = C_K\left[\sqrt{\frac{d + \log(2/\delta)}{N}} + \frac{d + \log(2/\delta)}{N}\right].$$

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Consequently, if

$$\gamma_{\mathrm{ff}} \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}} \le \frac{\mathcal{T}_k^\star}{4},$$

then the top dyad $\hat{u}_k\hat{v}_k^\top$ of $\widetilde{A}_k$ satisfies

$$\left\|\hat{u}_k\hat{v}_k^\top - u_{1,k}v_{1,k}^\top\right\|_{\mathrm{F}} \le C\frac{\gamma_{\mathrm{ff}} \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}}{\mathcal{T}_k^\star}.$$

Moreover, if $U_k = s_k\hat{u}_k\hat{v}_k^\top$ and $\varepsilon_{k,\mathrm{amp}} = \left|s_k - \sigma_1(\widetilde{A}_k)\right|$, then

$$\|U_k - U_k^\star\|_{\mathrm{F}} \le \varepsilon_{k,\mathrm{amp}} + C'\frac{\gamma_{\mathrm{ff}} \left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}}{\mathcal{T}_k^\star}\sigma_1(\Delta_k^\star).$$

D.4 Proofs of the bridge propositions

Proof of Proposition D.1. The squared-loss objective can be written as

$$\begin{aligned}\mathcal{L}_{\mathrm{ff}}(\mathbf{W}) &= \frac{1}{2}\mathbb{E}\,\mathrm{tr}\left[(\mathbf{W} - \mathbf{W}_0 - \Delta)^\top zz^\top(\mathbf{W} - \mathbf{W}_0 - \Delta)\right] \\ &= \frac{1}{2}\,\mathrm{tr}\left[(\mathbf{W} - \mathbf{W}_0 - \Delta)^\top\Sigma(\mathbf{W} - \mathbf{W}_0 - \Delta)\right].\end{aligned}$$

Differentiating gives

$$\nabla\mathcal{L}_{\mathrm{ff}}(\mathbf{W}) = \Sigma(\mathbf{W} - \mathbf{W}_0 - \Delta).$$

At stage k, $\mathbf{W}^{\langle k\rangle} - \mathbf{W}_0 - \Delta = -R_k^W$, and hence $-\nabla\mathcal{L}_{\mathrm{ff}}(\mathbf{W}^{\langle k\rangle}) = \Sigma R_k^W$. The empirical identity is identical with $\widehat{\Sigma}_{\mathcal{B}} = |\mathcal{B}|^{-1}\sum_{z_i\in\mathcal{B}} z_i z_i^\top$ in place of Σ. Therefore

$$\left\|\Sigma R_k^W - R_k^W\right\|_{\mathrm{F}} = \left\|(\Sigma - I)R_k^W\right\|_{\mathrm{F}} \le \|\Sigma - I\|_2 \left\|R_k^W\right\|_{\mathrm{F}},$$

and the same deterministic bound holds with $\widehat{\Sigma}_{\mathcal{B}}$.

When $\Sigma = I$, the population negative gradient is exactly R_k^W. If $\Delta = \sum_{j\ge1}\sigma_j u_j v_j^\top$ is an SVD, then stage 1 extracts $u_1v_1^\top$ with amplitude σ_1. After subtracting this component, the residual is $\sum_{j\ge2}\sigma_j u_j v_j^\top$, so induction gives the first r SVD components. The final parameter error is therefore the oracle tail. For the same selected SVD dyads $D_j = u_j v_j^\top$, joint scalar refitting solves

$$\min_{c\in\mathbb{R}^r}\frac{1}{2}\left\|\Delta - \sum_{j=1}^r c_j D_j\right\|_{\mathrm{F}}^2.$$

The dyads are orthonormal in Frobenius inner product, so the unique minimizer on the selected span is $c_j = \langle\Delta, D_j\rangle = \sigma_j$ up to arbitrary choices inside tied singular subspaces. Finally, for any fixed dyads and any fixed stagewise scalar values, those values are feasible for the joint least-squares problem; the optimizer over the same fixed span cannot have larger squared loss.

Proof of Proposition D.2. For isotropic sub-Gaussian features, the standard sample-covariance bound [see, e.g., Vershynin, 2018] gives, with probability at least $1-\delta$,

$$\left\|\widehat{\Sigma}_{\mathcal{B}_k} - I\right\|_2 \le C_K\left[\sqrt{\frac{d + \log(2/\delta)}{N}} + \frac{d + \log(2/\delta)}{N}\right] = \gamma_{\mathrm{ff}},$$

where C_K depends only on the sub-Gaussian norm bound. Since $\widetilde{A}_k = \widehat{\Sigma}_{\mathcal{B}_k}R_k^W$,

$$\left\|\widetilde{A}_k - R_k^W\right\|_{\mathrm{F}} = \left\|(\widehat{\Sigma}_{\mathcal{B}_k} - I)R_k^W\right\|_{\mathrm{F}} \le \left\|\widehat{\Sigma}_{\mathcal{B}_k} - I\right\|_2\left\|R_k^W\right\|_{\mathrm{F}} \le \gamma_{\mathrm{ff}}\left\|R_k^W\right\|_{\mathrm{F}}.$$

Therefore

$$\left\|\widetilde{A}_k - \Delta_k^\star\right\|_{\mathrm{F}} \le \gamma_{\mathrm{ff}}\left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}.$$

If this quantity is at most $\mathcal{T}_k^\star/4$, Wedin's singular-vector perturbation bound applied to $\widetilde{A}_k$ and $\Delta_k^\star$ gives the displayed dyad estimate, up to a universal numerical constant. Weyl's inequality gives

$$\left|\sigma_1(\widetilde{A}_k) - \sigma_1(\Delta_k^\star)\right| \le \gamma_{\mathrm{ff}}\left\|R_k^W\right\|_{\mathrm{F}} + \left\|R_k^W - \Delta_k^\star\right\|_{\mathrm{F}}.$$

Combining this amplitude perturbation with the dyad perturbation exactly as in the proof of Proposition 4.5 yields the stated bound on $\|U_k - U_k^\star\|_{\mathrm{F}}$.

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D.5 Frozen-feature bridge experiments

We use the frozen-feature bridge in two complementary ways. In a *synthetic-feature bridge* we replace real labels with a known low-rank Leaky-ReLU teacher and measure the same per-stage diagnostics as in the synthetic suite, since the teacher shift is known and we can compute oracle residuals, gaps, dyad error, sector angles, and the perturbation ratio of Proposition 4.5. In a *real-label bridge* we fit a single classifier head over the same frozen features and report the rank/active-state frontier that motivates the procedure. In both settings the Gaussian-input assumption no longer holds, so the relevant local statement is the squared-loss bridge of Proposition D.1. That proposition predicts the population-gradient surrogate is exactly ΣR, with empirical perturbation bounded by $\|\widehat{\Sigma}_{\mathcal{B}} - I\|_2 \|R\|_{\mathrm{F}}$. Whitening should therefore drive the perturbation ratio down through $\widehat{\Sigma} \to I$, and we use this as our reading of the bridge.

Datasets and backbones. We use CIFAR-10 and CIFAR-100 [Krizhevsky and Hinton, 2009] and Flowers102 [Nilsback and Zisserman, 2008]. We pass each image through an ImageNet-pretrained ResNet-18 [He et al., 2016] or MobileNetV3-Small [Howard et al., 2019] and freeze the backbone. We sample at most 12000 training images and 3000 test images at a fixed feature seed, project the features to dimension 256, and standardise them on the training split. By default we then apply covariance whitening with ridge $5 \cdot 10^{-3}$. We do not update any backbone parameter.

Real-label adaptation. For the real-label bridge we first fit a frozen linear head on 4000 labelled examples for 500 Adam steps at learning rate $5 \cdot 10^{-2}$. We then adapt only the head matrix. Locked methods use fresh stage batches of size 4096. Scalar Adam takes 120 steps at learning rate $5 \cdot 10^{-2}$. The free-factor sequential baseline shares the SVD initialisation but optimises both factors for 160 Adam steps at learning rate $3 \cdot 10^{-2}$. The simultaneous baseline optimises all factors jointly for 650 Adam steps at the same factor learning rate. We sweep final ranks $\{1, 2, 4, 6\}$, run each frontier with cross-entropy and with a one-vs-rest squared-loss head initialised by ridge regression, and use five seeds 11–15. The squared-loss head is closer to the bridge proposition because the loss is quadratic in the head output.

We measure active Adam-state memory as a function of trainable adaptation parameters: zero for locked spectral extraction, one scalar for locked scalar refinement, r scalars for fixed-direction refitting, one rank-1 factor pair for free-factor sequential training, and rank-r factors for simultaneous training. We also record active optimizer-state size and adaptation FLOPs. At rank 6 with cross-entropy, scalar refitting uses $4.8 \cdot 10^{-5}$ MB of active Adam state for the trainable scalars, whereas simultaneous training uses 0.0128, 0.0171, and 0.0172 MB on CIFAR-10, CIFAR-100, and Flowers102 respectively. On CIFAR-10 the one-run scalar-refit frontier reaches $75.1\% \pm 1.1\%$ at rank 6 after 22.2 head-adaptation GFLOPs, while separately training simultaneous ranks $1, 2, 4, 6$ reaches $77.6\% \pm 0.2\%$ at rank 6 after 53.9 cumulative head-adaptation GFLOPs.

Synthetic-feature bridge. We replace CIFAR labels with a known Leaky-ReLU teacher of 24 outputs, rank 4, $\alpha = 0.05$, and shift Frobenius norm 0.65. Because the teacher is known we report oracle residuals, residual gaps, dyad error, sector angles, and the perturbation ratio of Proposition 4.5. We run the diagnostic on raw, standardised, PCA-projected, and whitened features (ridge values $\{10^{-1}, 10^{-2}, 5 \cdot 10^{-3}\}$) and on a randomly rotated whitened control, and we repeat it with the MobileNetV3-Small backbone. The Gaussian theorem does not apply on these features. We use the diagnostic to measure how close the bridge is to the local geometry needed by the theorem before considering larger fine-tuning runs.

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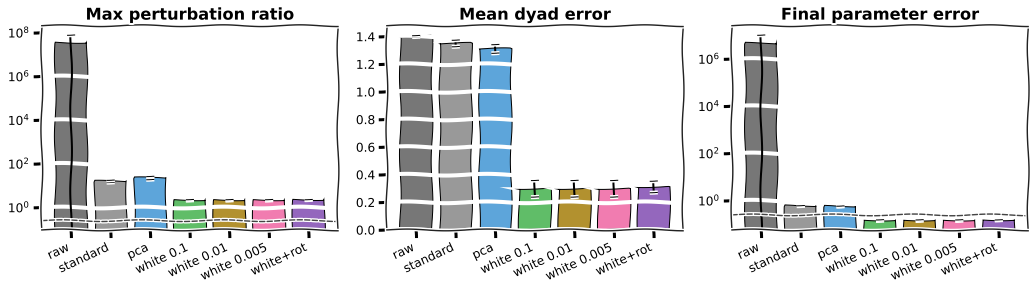


Figure 12: Frozen-feature bridge: effect of preprocessing on a known low-rank teacher. Two pretrained backbones (ResNet-18, MobileNetV3-Small) with raw, standardized, PCA-projected, and whitened features (three ridge values). Whitening is the preprocessing that brings $\widehat{\Sigma}$ closest to I; consistent with the $\|\Sigma - I\|_2\|R\|_{\mathrm{F}}$ perturbation bound of Proposition D.1, it improves the perturbation ratio, top dyad error, and final parameter error compared with the alternatives. The Gaussian sufficient threshold is not reached on either backbone, so the figure illustrates the bridge mechanism rather than verifying the Gaussian theorem on real features.

Whitening drives the bridge perturbation ratio down (Figure 12). Figure 12 reports the per-stage diagnostic across preprocessing choices on both backbones. Whitening is the only preprocessing that brings the perturbation ratio close to the bridge threshold. Even after whitening the sufficient Gaussian threshold is not reached on either backbone, so the figure illustrates the bridge mechanism but does not verify the Gaussian theorem on real features.

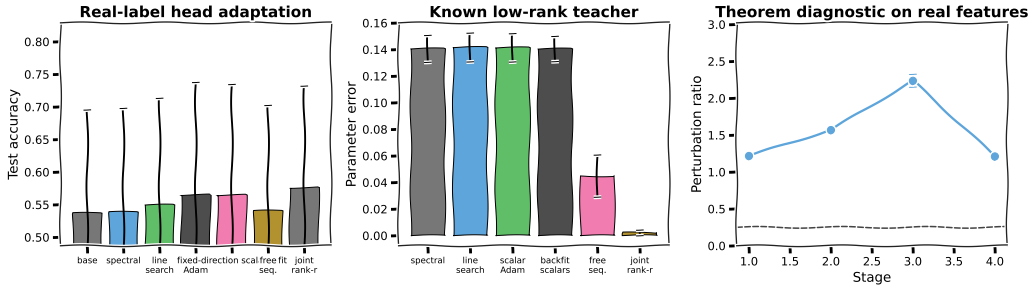


Figure 13: Frozen-feature CIFAR-10 bridges. Left: real-label adaptation of a single classifier head over frozen pretrained features. Middle: parameter recovery for a known low-rank Leaky-ReLU teacher on the same features. Right: empirical perturbation ratio for the locked scalar method on the synthetic-feature bridge; the dashed line is the sufficient Gaussian threshold underlying Proposition 4.5. The threshold is not met on these features even after whitening, so the figure is bridge evidence for the residual-surrogate mechanism (Proposition D.1/Proposition D.2) rather than a verification of the Gaussian Leaky-ReLU theorem on real data.

Three views of the CIFAR-10 bridge (Figure 13). Figure 13 aggregates three views of the CIFAR-10 bridge: real-label head adaptation, parameter recovery against the known teacher, and the per-stage perturbation ratio from the synthetic-feature bridge. None of the panels meets the sufficient threshold, and we read the figure as bridge evidence for the residual-surrogate mechanism rather than as a verification of the Gaussian theorem.

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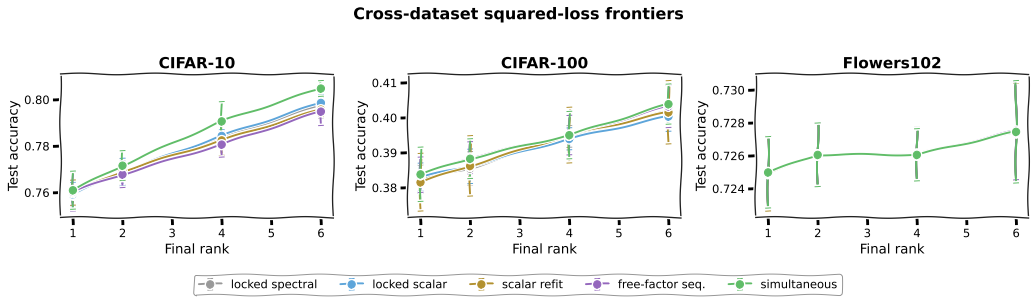


Figure 14: Squared-loss one-vs-rest head frontiers on frozen ResNet-18 features. This bridge is closer to the analyzed setting than cross-entropy: the adapted object is a single matrix and the loss is quadratic in the head outputs, matching Proposition D.1. At rank 6, fixed-direction scalar refitting nearly matches simultaneous rank-r training on all three datasets. The figure is empirical evidence for the active-state/rank-frontier tradeoff, not a verification of the Gaussian theorem in deep settings.

Frontiers under squared loss (Figure 14). Figure 14 reports rank/accuracy frontiers under the one-vs-rest squared-loss head, which is the loss most aligned with Proposition D.1. At rank 6, fixed-direction scalar refitting nearly matches simultaneous rank-r training on all three datasets. This is the closest the bridge gets to the analysed setting and is empirical evidence for the active-state/rank-frontier tradeoff, not a verification of the Gaussian theorem.

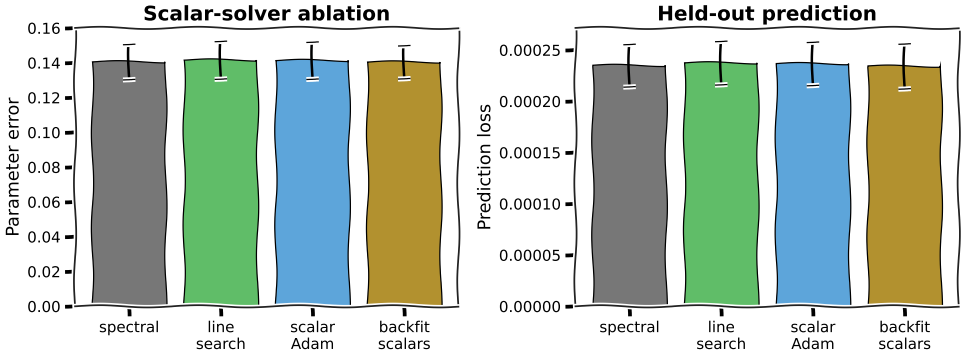


Figure 15: Scalar-solver ablation on the synthetic-feature bridge. Spectral amplitude, numerical line search, scalar Adam, and scalar backfitting yield similar final error after whitening; consistent with the per-stage slack $\varepsilon_{k,\mathrm{amp}}$ in Proposition 4.5 being dominated by the recovered direction rather than the choice of one-dimensional solver.

Scalar-solver ablation under the bridge (Figure 15). In Figure 15 we repeat the synthetic scalar-solver ablation on the synthetic-feature bridge. Spectral amplitude, line search, scalar Adam, and scalar backfitting again give similar final errors after whitening, in line with the per-stage slack $\varepsilon_{k,\mathrm{amp}}$ being dominated by the recovered direction.

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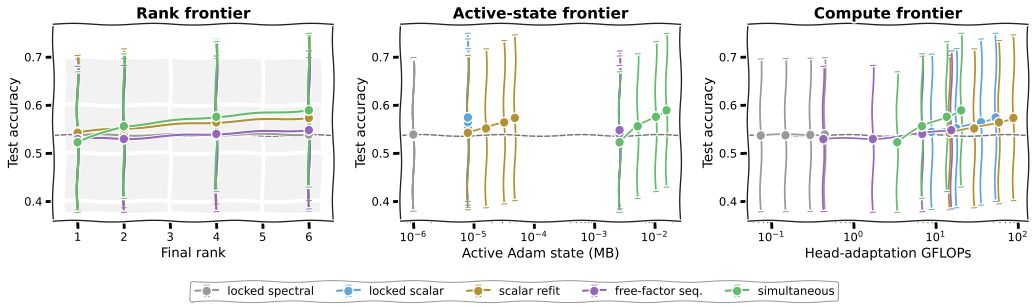


Figure 16: Frozen-feature CIFAR-10 rank/active-state frontier. Adapted object: one classifier head over frozen ImageNet-pretrained ResNet-18 features, with the base head fit on 4000 labels. Fixed-direction scalar refitting is a bridge for the analyzed setting (Proposition D.1/Proposition D.2) and produces an entire rank frontier in a single run with bounded active optimizer state. Simultaneous rank-r training is an empirical baseline outside the theorem: it is more accurate at each rank but uses rank-r active optimizer state and must be retrained per target rank.

Cross-entropy active-state frontier (Figure 16). Figure 16 reports the cross-entropy rank/active-state frontier on CIFAR-10 with the same 4000-example base head. Fixed-direction scalar refitting yields the entire rank frontier in a single run with bounded active optimizer state. Simultaneous rank-r training is more accurate at each rank but uses rank-r active state and must be retrained for every target rank. Cross-entropy is outside the squared-loss bridge of Proposition D.1, so we report this only for empirical context.

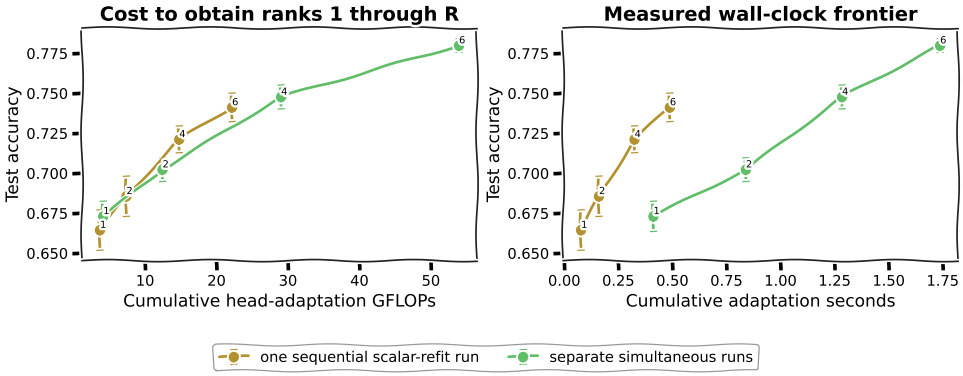


Figure 17: Anytime rank frontier on CIFAR-10 with the same 4000-example base head as Figure 16. A single fixed-direction scalar-refit run yields ranks $1, 2, 4, 6$ along one path; the simultaneous baseline is more accurate at each rank but requires separately training a model per target rank. The figure illustrates the active-state/anytime tradeoff implied by sequential rank-one extraction; the underlying Gaussian theorem is not exercised here.

Anytime frontier from a single run (Figure 17). Figure 17 highlights the anytime nature of the procedure: a single fixed-direction scalar-refit run produces ranks $1, 2, 4, 6$ along one path. The simultaneous baseline is more accurate at each rank but requires a separate model per target rank.

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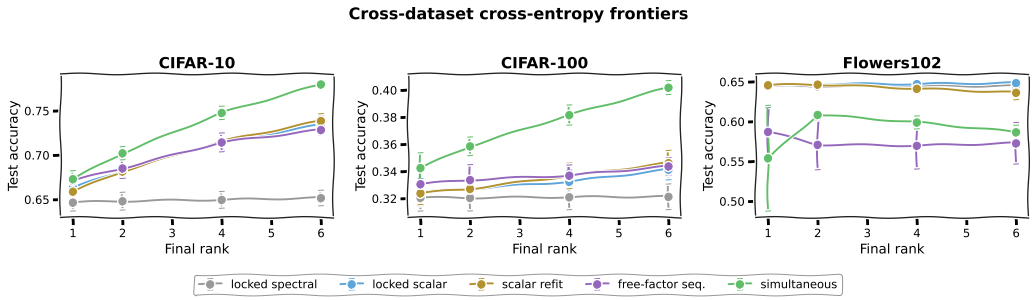


Figure 18: Cross-dataset cross-entropy head frontiers on frozen ResNet-18 features. CIFAR-10 and CIFAR-100 show a useful active-state tradeoff for fixed-direction scalar refitting; Flowers102 is mostly saturated by the frozen base head and is included as a control where sequential updates do not help. Cross-entropy is outside the squared-loss bridge of Proposition D.1, so this panel is empirical evidence for the active-state tradeoff rather than a verification of the bridge.

Cross-dataset frontiers (Figure 18). In Figure 18 we extend the cross-entropy frontier to CIFAR-10, CIFAR-100, and Flowers102. CIFAR-10 and CIFAR-100 show a useful active-state tradeoff for fixed-direction scalar refitting. Flowers102 is mostly saturated by the frozen base head, and we include it as a control where sequential updates do not help.

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